

ALIBABA CLOUD INFRASTRUCTURE GUIDE FOR AZURE PROFESSIONALS

CLOUD MIGRATION
WHITEPAPER SERIES

 Alibaba Cloud

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TARGET AUDIENCE

This guide is intended to provide Azure professionals with methodologies and best practices to understand Alibaba Cloud infrastructure, regardless of whether you are planning a multi-cloud solution with Azure and Alibaba Cloud or migrating to Alibaba Cloud.





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1. WHY MAKE THE MOVE?

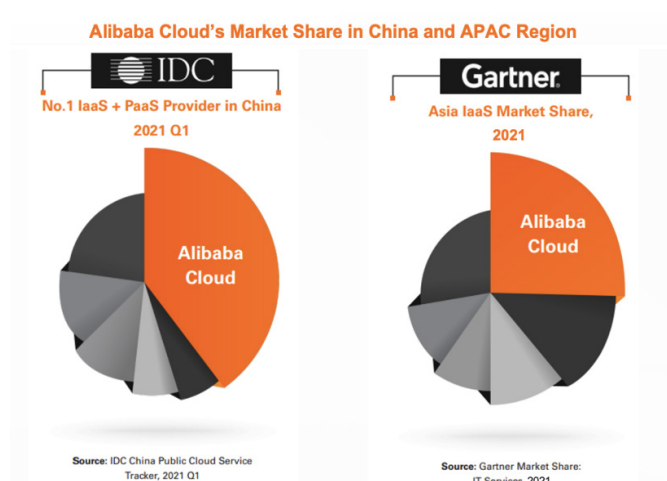
Here are some key considerations that may drive customers to make the move from Azure to Alibaba Cloud or plan a multi-cloud solution with Azure and Alibaba Cloud.

1.1 ALIBABA CLOUD MARKET PRESENCE

Alibaba Cloud, founded in 2009, is a global leader in cloud computing and artificial intelligence, providing services to thousands of enterprises, developers, and government organizations in more than 200 countries and regions. Committed to the success of its customers, Alibaba Cloud provides reliable and secure cloud computing and data processing capabilities as a part of its online solutions. In January 2017, Alibaba Cloud became the official cloud services partner of the Olympics.

According to the latest data released by IDC on China's public cloud market in the first quarter of 2021, the IAAs + PAAS market reached US \$4.632 billion (30.1 billion yuan) in the quarter, of which Alibaba cloud ranked first with a market share of 40%.

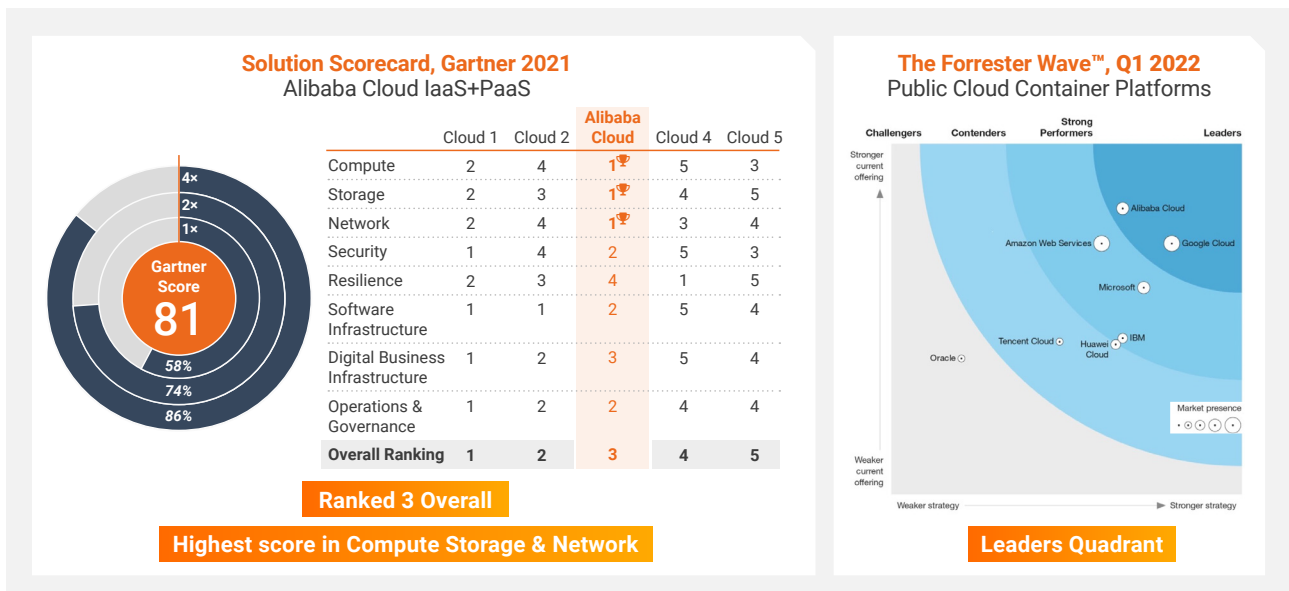
For the fourth year in a row, Alibaba Cloud is ranked as the third largest Infrastructure-as-a-Service (IaaS) provider in the world and the biggest in Asia Pacific in 2021 by revenue. The latest Gartner® report, Market Share: IT Services, 2021, showed Alibaba's global market share expanded to 9.55%, while in Asia Pacific the company has achieved 25.5%.



1.2 THIRD-PARTY EVALUATION

In the 2021 Gartner Solution Scorecard for IaaS + PaaS, Alibaba Cloud achieved a score of 81 out of 100. Alibaba Cloud has been included in this report for 3 years in a row and is also the only Chinese cloud vendor selected for this report so far. Alibaba Cloud received the highest scores in the compute, storage, and network categories respectively, and the highest score in security in the required criteria.

Also, Alibaba Cloud was named a Leader in The Forrester Wave™: Public Cloud Container Platforms, Q1 2022. Forrester views Alibaba Cloud as a leader in the public cloud container platforms market, and Alibaba Cloud is a great fit for global enterprises targeting China and Asia Pacific markets, or local firms going abroad that seek comprehensive cloud-native capabilities for digital transformation with proven experience.



There are also key findings and recommendations from 2021 Gartner Solution Scorecard for IaaS + PaaS about Alibaba Cloud:

"It is suitable for *enterprise-class* production use."

"Significant differentiators of its own, particularly in data and *artificial intelligence (AI)*-/*machine learning (ML)*-oriented services."

"Select Alibaba Cloud as your *primary strategic* IaaS+PaaS provider if you are an *APAC-based* business ..."

"Evaluate Alibaba Cloud as a *strategic* IaaS+PaaS provider within a *multi-cloud strategy* ..."

"Assess Alibaba Cloud as a potential *tactical* IaaS+PaaS provider for workloads that you would prefer to host in *China* ..."



1.3 GO CHINA & GO ASIA

Alibaba Cloud was born and raised in China and understands the complexities and opportunities of China's market, which led to the creation of the outstanding "Go China" solutions. Learn more in [China Gateway](#). As a native Asian company and Asia's No. 1 IaaS service provider recognized by Gartner¹, Alibaba Cloud can accelerate your success in Asia. Learn more with [Asia Accelerator](#).

- » [ICP Filing Assistant](#)
- » [MLPS 2.0 Compliance](#)
- » [Connecting Your Business with Asia](#)
- » [Safeguard Your Business in Asia](#)
- » [Global ISVs China Onboarding](#)

1.4 ECOSYSTEM

Alibaba Cloud has more than 10,000 global partners and connects with more than 10 business units of Alibaba Group to offer better service to customers, covering more than 200 industry solutions, including finance, retail, e-commerce, blockchain, media, and more.

As with Azure, Alibaba Cloud provides a comprehensive list of cloud-native products, including containers, APIs, CI/CD technology, database-as-a-service, etc. Compared with Azure, Alibaba Cloud offers better support for non-MS products, such as MySQL, and MongoDB.

1.5 COST EFFICIENCY

One of the major factors driving a migration from Azure is cost efficiency. For most of the products on Azure, customers can find similar products on Alibaba Cloud with competitive prices, good performance, and high availability. In most situations, when we compare prices for cloud products of similar configuration, Alibaba Cloud is more attractive than Azure, and Azure instances tend to get costlier as the size increases.

Furthermore, Alibaba Cloud offers flexible, easy-to-use, and cost-effective resource packages to further save your cost, such as monthly subscriptions for many of the Alibaba Cloud products, the Storage Capacity Unit (SCU) for storage, Data Transfer Plan for network traffic, and 95 percentile billing for CEN, etc.

- » [Excellent Computing Powered by X-Dragon Architecture](#)
- » [Industry-Leading ECS Stability Proven Database Reliability and Scalability](#)

¹ Gartner Market Share: IT Services, Worldwide, 2019, 13 April 2020

- » [Up to 76% PAYG Cost Reduction with Savings Plans](#)
- » [Flexible Pricing Options](#)

Especially, when you want to deploy your business in a Microsoft software-based environment, Microsoft licenses can be provided through Alibaba Cloud. You can use one of the following methods to use a Microsoft license in Alibaba Cloud:

USE MICROSOFT LICENSES PROVIDED BY ALIBABA CLOUD

Alibaba Cloud is licensed by the Services Provider License Agreement (SPLA) to provide Microsoft licenses. Alibaba Cloud Elastic Compute Service (ECS) and ApsaraDB RDS provide software licensed by Microsoft, such as Windows Server images and ApsaraDB RDS for SQL Server databases. Client Access Licenses (CALs) are not required for Alibaba Cloud ECS instances that run Windows Server images. By default, each Windows Server instance has two connections that are used only for management purposes. If you need more connections or want to use the connections for non-management purposes, an extra Remote Desktop Services CAL (RDS CAL) is required.

You can select an appropriate Windows Server image or database type when you create an ECS instance. The following table lists the versions of Windows Server images and SQL Server versions provided by Alibaba Cloud.

Windows Server version	SQL Server version
Windows Server 2022 Datacenter 64-bit (Simplified Chinese)	SQL Server 2019 Standard on Windows Server 2016
Windows Server 2022 Datacenter 64-bit (English)	SQL Server 2017 Web on Windows Server 2016
Windows Server 2019 Datacenter with Containers 64-bit (Simplified Chinese)	SQL Server 2017 Enterprise on Windows Server 2016
Windows Server 2019 Datacenter with Containers 64-bit (English)	SQL Server 2017 Standard on Windows Server 2016
Windows Server 2019 Datacenter 64-bit (Simplified Chinese)	SQL Server 2016 Enterprise on Windows Server 2016
Windows Server 2019 Datacenter 64-bit (English)	SQL Server 2016 Standard on Windows Server 2016
Windows Server 2016 Datacenter 64-bit (Simplified Chinese)	SQL Server 2014 Web on Windows Server 2016
Windows Server 2016 Datacenter 64-bit (English)	SQL Server 2014 Enterprise on Windows Server 2016

Windows Server version	SQL Server version
Windows Server 2012 R2 Datacenter 64-bit (Simplified Chinese)	SQL Server 2014 Standard on Windows Server 2016
Windows Server 2012 R2 Datacenter 64-bit (English)	SQL Server 2012 Web on Windows Server 2012
Windows Server Version **** Datacenter 64-bit (Simplified Chinese)	SQL Server 2012 SP3 Enterprise on Windows Server 2012 R2
Windows Server Version **** Datacenter 64-bit (English)	SQL Server 2012 SP3 Standard on Windows Server 2012 R2

Note On January 14, 2020, Microsoft stopped providing support for Windows Server 2008 and Windows Server 2008 R2 operating systems. In light of this, Alibaba Cloud no longer provides technical support for ECS instances that use these operating systems. If you have ECS instances that use these operating systems, upgrade them to Windows Server 2012 or later at your earliest opportunity.

BRING YOUR OWN LICENSE

If you have already purchased Microsoft licenses, you can continue to use them by bringing them to Alibaba Cloud. Check whether your Microsoft license has Software Assurance coverage. For more information, visit the [official Microsoft website](#). You can perform one of the following operations based on whether your Microsoft license has Software Assurance coverage.

Note You can import a BYOL image to Alibaba Cloud and use this image to create ECS instances. Set License Type to BYOL when you import a BYOL image.

- » Microsoft licenses without Software Assurance coverage. If your Microsoft license does not have Software Assurance coverage, you can use ECS bare metal instances or dedicated hosts. The resources offered by these instances or hosts are exclusive for your use, so you can use your own Microsoft licenses even though these licenses do not have Software Assurance coverage.

Important To meet compliance requirements, you must have obtained your license before October 1, 2019 and the operating system versions must have been released before October 1, 2019. All Microsoft software covered by the Microsoft Product Terms can be deployed on ECS bare metal instances or dedicated hosts. Examples include:

- » Microsoft Windows Server
- » Microsoft Exchange Server
- » Microsoft SharePoint Server
- » Microsoft SQL Server



- » *Microsoft System Center*
 - » *Microsoft Dynamics*
 - » *MSDN*
 - » *Microsoft Remote Desktop Services (user CALs)*
- » Microsoft licenses with Software Assurance coverage. You can migrate your Microsoft license to Alibaba Cloud and continue to use it by using the Software Assurance and Microsoft Volume Licensing programs. For software covered by the license terms of valid Microsoft products, you can apply for License Mobility through Software Assurance before you migrate services to ECS.

The following list describes eligible Microsoft applications and services:

- » *Microsoft Exchange Server*
- » *Microsoft SharePoint Server*
- » *Microsoft SQL Server*
- » *Microsoft System Center*
- » *Microsoft Dynamics*
- » *Microsoft BizTalk Server*
- » *Microsoft Remote Desktop Services (user CALs)*
- » *Microsoft Skype for Business Server*

1.6 MULTI-CLOUD / HYBRID CLOUD

Alibaba Cloud is the pilot of hybrid clouds and has supported many customer success stories across many industries. Inherited from Alibaba Cloud public cloud, Alibaba Cloud Hybrid Cloud is equipped with over 50 full-stack product portfolios and solutions, enabling you to practice hybrid cloud in various business scenarios.

- » [Extend Your Business Across On-Premise and the Cloud](#)
- » [Things to Know About a Multi-Cloud Solution](#)
- » [Hybrid Cloud Products](#)



1.7 SECURITY AND COMPLIANCE

The security process of Alibaba Cloud has been recognized by authorities inside and outside mainland China. By leveraging its years of expertise in defending against internet security threats, Alibaba Cloud provides security protection for the cloud platform and integrates multiple compliance standards into the internal control and product design of the cloud platform.

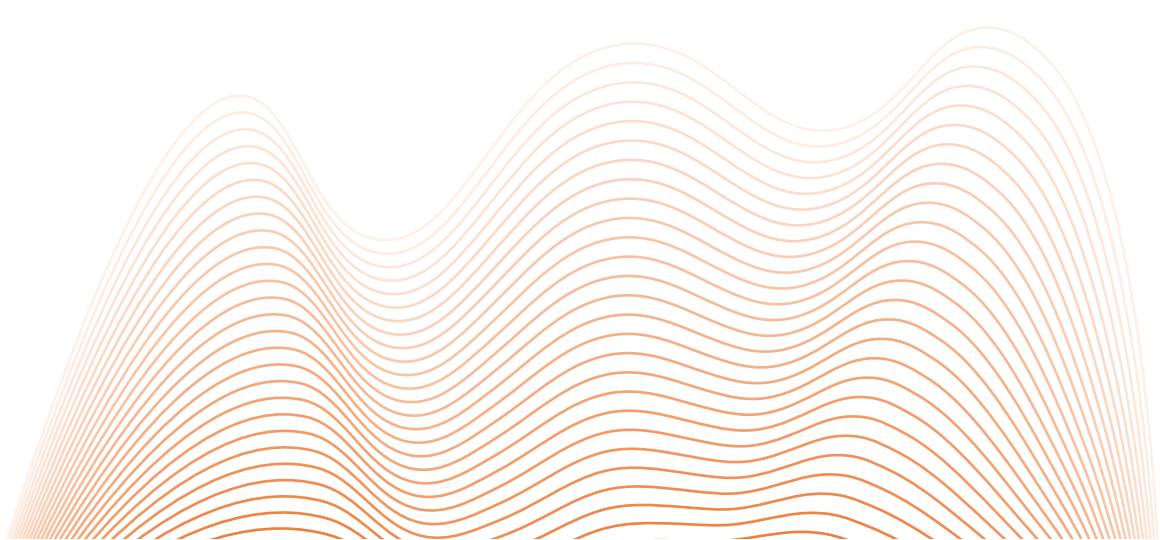
User data security on the cloud is the lifeline of users and one of the most important representational attributes of the overall cloud security capability. In July 2015, Alibaba Cloud initiated the first Data Protection Proposal among cloud service providers in China. This public proposal appeals that the ownership of data of all developers, companies, governments, and social institutions on the cloud computing platforms belongs only to the users. Cloud computing platforms cannot use the data for other purposes. Cloud service providers have the responsibilities and obligation to help users ensure the confidentiality, integrity, and availability of their data.

- » [Security and Privacy Compliance](#)
- » [Information Security](#)
- » [Data Security](#)

1.8 INSTANT SUPPORT

Alibaba Cloud offers highly flexible support services tailored to meet your exact needs. A dedicated technical account manager (TAM) will be assigned, and a dedicated DingTalk instant messaging group will be created to support enterprise customers.

- » [After-Sales Support](#)
- » [Partner Support Plan](#)



2. SERVICES COMPARISON BETWEEN AZURE AND ALIBABA CLOUD

By March 2023, Alibaba Cloud has 86 zones strategically located across 28 regions in the world, providing reliable computing support for billions of customers. Alibaba Cloud has the most data centers (76 zones located across 23 regions) in the Asia Pacific area – more than any other public cloud provider. Please visit the Alibaba Cloud website to check our [Data Centers](#) around the world.

Area	Country	City	Region Id	Number of Zones
Asia Pacific (23)	China	Beijing	cn-beijing	12
	China	Chengdu	cn-chengdu	2
	China	Fuzhou	cn-fuzhou	1
	China	Guangzhou	cn-guangzhou	2
	China	Hangzhou	cn-hangzhou	8
	China	Heyuan	cn-heyuan	2
	China	Hong Kong	cn-hongkong	3
	China	Huhehot	cn-huhehaote	2
	China	Nanjing	cn-nanjing	1
	China	Qingdao	cn-qingdao	2
	China	Shanghai	cn-shanghai	11
	China	Shenzhen	cn-shenzhen	6
	China	Ulanqab	cn-wulanchabu	3
	China	Zhangjiakou	cn-zhangjiakou	3
	Japan	Tokyo	ap-northeast-1	3
	Korea	Seoul	ap-northeast-2	1
	India	Mumbai	ap-south-1	2
	Singapore	Singapore	ap-southeast-1	3

Area	Country	City	Region Id	Number of Zones
Asia Pacific (23) – continued	Australia	Sydney	ap-southeast-2	2
	Malaysia	Kuala Lumpur	ap-southeast-3	2
	Indonesia	Jakarta	ap-southeast-5	3
	Philippines	Manila	ap-southeast-6	1
	Thailand	Bangkok	ap-southeast-7	1
Europe & Americas (4)	Germany	Frankfurt	eu-central-1	3
	UK	London	eu-west-1	2
	US	Silicon Valley	us-west-1	2
	US	Virginia	us-east-1	2
Middle East (1)	UAE	Dubai	me-east-1	1

Alibaba Cloud offers comprehensive services. For most of the services on Azure, customers can find similar services on Alibaba Cloud. Below is a rough comparison of services offered by Azure and Alibaba Cloud.

Please note that not every Azure service or Alibaba Cloud service is listed, and not every matched service has exact feature-for-feature parity.

2.1 ELASTIC COMPUTING

2.1.1 CLOUD SERVER

Azure Product	Alibaba Cloud Product	Description
Virtual Machines	Elastic Compute Service (ECS)	Elastic and secure virtual cloud servers to cater all your cloud hosting needs.
Bare Metal Infrastructure	ECS Bare Metal Instance (EBM)	An elastic and horizontally scalable high-performance computing service providing the same computing performance as traditional physical servers including physical isolation.
App Service Environment	Simple Application Server	A single server-based service for application deployment, security management, O&M monitoring, and more.
GPU Optimized VMs	Elastic GPU Service (EGS)	Powerful parallel computing capabilities based on GPU technology.
Dedicated Host	Dedicated Host (DDH)	A dedicated resource pool that offers flexible and compliant deployment options to grant enterprise users more control over their cloud infrastructure.
Virtual Desktop	Elastic Desktop Service (EDS)	A convenient and secure cloud-based Desktop-as-a-Service (DaaS) solution.

2.1.2 ELASTIC ORCHESTRATION

Azure Product	Alibaba Cloud Product	Description
Autoscale	Auto Scaling	Automatically adjusts computing resources based on your business cycle.
Resource Manager	Resource Orchestration Service (ROS)	A fully managed service for cloud computing resource orchestration and automated deployment.
Azure Automation	Operation Orchestration Service (OOS)	Automated operations and maintenance (O&M) service.

2.1.3 HIGH-PERFORMANCE COMPUTING

Azure Product	Alibaba Cloud Product	Description
High-Performance Computing	Elastic High Performance Computing (E-HPC)	An HPCaaS cloud platform providing an all-in-one high-performance public computing service.
Azure Batch	Batch Compute	Resource management and task scheduling for large-scale batch processing.

2.2 NETWORKING

2.2.1 CLOUD NETWORK

Azure Product	Alibaba Cloud Product	Description
Virtual Network	Virtual Private Cloud (VPC)	A virtual private cloud service that provides an isolated cloud network to operate resources in a secure environment.
Load Balancer	Network Load Balancer (NLB)	Network Load Balancer (NLB) is a Layer 4 load balancing service intended for the Internet of Everything (IoE) era. NLB offers ultra-high performance and can automatically scale on demand. An NLB instance supports up to 100 million concurrent connections, which is ideal for services that require high concurrency.
Private Link	Private Link	Connect your VPCs to services in other VPCs through secure, reliable, and private connections. Private Link simplifies network architecture and reduces security risks because network traffic does not traverse the internet.



Azure Product	Alibaba Cloud Product	Description
Virtual Network Peering	VPC Peering Connections	A virtual private cloud (VPC) peering connection is a private network connection between two VPCs. You can enable two VPCs to communicate with each other by establishing a VPC peering connection. You can create a VPC peering connection between VPCs that belong to the same Alibaba Cloud account (same-account) or different accounts (cross-account). You can also create VPC peering connections between VPCs that belong to the same region (intra-region) or different regions (inter-region).
Virtual Network NAT	NAT Gateway	A NAT gateway enables multiple instances within a virtual private cloud (VPC) to communicate with the internet. Custom SNAT and DNAT rules can be created to help you use network resources in a flexible manner.
Virtual Network IP Services	Elastic IP Address (EIP)	EIPs are independent public IP address resources, which allow you to decouple public IP addresses from ECS instances and facilitate management.
Traffic Manager	Global Traffic Manager (GTM)	Allows you to access the nearest node based on the Domain Name System (DNS) architecture. With built-in health checks, you can easily monitor your applications and run a failover when a failure occurs on a node.
Virtual Network Manager	Network Intelligence Service (NIS)	Network Intelligence Service (NIS) is an intelligent self-service platform that allows you to design, deploy, and maintain networks for your websites. NIS provides statistics that can help you design your network and troubleshoot network issues.
Azure DNS	Alibaba Cloud DNS PrivateZone	Alibaba Cloud DNS PrivateZone is a private domain name resolution and management service based on Virtual Private Cloud (VPC). You can use Alibaba Cloud DNS PrivateZone to resolve private domain names to IP addresses in one or more custom VPCs.
Application Gateway	Application Load Balancer (ALB)	Application Load Balancer (ALB) is an Alibaba Cloud service that runs at the application layer and is optimized to balance traffic over HTTP, HTTPS, and Quick UDP Internet Connections (QUIC). ALB is highly elastic and can scale on demand to process large volumes of Layer 7 traffic.

2.2.2 CROSS-REGION NETWORK

Azure Product	Alibaba Cloud Product	Description
Azure Virtual WAN	Cloud Enterprise Network (CEN)	A global network for rapidly building a distributed business system and hybrid cloud to help users create a network with enterprise-level scalability and the communication capabilities of a cloud network.
Traffic Manager	Global Accelerator (GA)	Provides network acceleration service for your internet-facing application globally with guaranteed bandwidth and high reliability.

2.2.3 HYBRID CLOUD NETWORK

Azure Product	Alibaba Cloud Product	Description
Azure VirtualWAN	Express Connect	A global network for rapidly building a distributed business system and hybrid cloud to help users create a network with enterprise-level scalability and the communication capabilities of a cloud network.
Traffic Manager	VPN Gateway	Provides network acceleration service for your internet-facing application globally with guaranteed bandwidth and high reliability.

2.3 STORAGE

2.3.1 DATA TRANSPORT

Azure Product	Alibaba Cloud Product	Description
Azure Data Box	Data Transport	An easy-to-use network service that enables high-bandwidth, reliable, secure, and private connections between different environments.

2.3.2 ESSENTIAL STORAGE SERVICE

Azure Product	Alibaba Cloud Product	Description
Azure Premium Storage	Elastic Block Storage (EBS)	Block-level data storage attached to ECS instances to achieve high performance, low latency, and high reliability.
Azure Blob Storage	Object Storage Service (OSS)	An encrypted and secure cloud storage service that can store, process, and access massive amounts of data from anywhere in the world.
Azure Files	File Storage NAS	Simple, scalable, on-demand, and reliable network-attached storage for use with ECS instances, HPC and Container Service.
Table Storage	Tablestore	A fully managed NoSQL cloud database service that enables the storage of a massive amount of structured and semi-structured data.



2.3.3 HYBRID CLOUD STORAGE

Azure Product	Alibaba Cloud Product	Description
Azure StorSimple	Cloud Storage Gateway (CSG)	Cloud Storage Gateway uses OSS for cloud-based storage at the back end, and supports standard file and block storage protocols in the industry.

2.3.4 STORAGE DATA SERVICE

Azure Product	Alibaba Cloud Product	Description
Event Hub, Log Analytics, Azure Monitor, Azure Kusto, App Logic	Log Service (SLS)	All-in-one service for log-type data.
Azure Backup	Hybrid Backup Recovery (HBR)	An enterprise-level data backup and migration service that is secure, efficient, cost-effective, and scalable. Hybrid Backup Recovery provides full protection for data stored in Alibaba Cloud and on-premises data centers.

2.4 DATABASE

2.4.1 DATA WAREHOUSE

Azure Product	Alibaba Cloud Product	Description
Azure Synapse Analytics	AnalyticDB for MySQL	A real-time data warehousing service that can process PB data with high concurrency and low latency.
Azure Synapse Analytics	AnalyticDB for PostgreSQL	An online MPP (Massively Parallel Processing) data warehousing service based on the open-source Greenplum Database. It provides an online expansion and performance monitoring service to free your team from complicated MPP cluster operations and management (O&M).
Azure Data Lake Analytics	Data Lake Analytics (DLA)	Alibaba Cloud Data Lake Analytics (DLA) is an interactive analytics service that utilizes serverless architecture. As a ready-to-use service, DLA does not require any prior setup of infrastructure or upfront management costs. You do not need to maintain instances in DLA, and service is billed based on actual use and needs. DLA uses SQL interfaces to interact with user service clients, which means it complies with standard SQL syntax and provides a variety of similar functions. DLA allows you to retrieve and analyze data from multiple data sources or locations such as OSS and Table Store for optimal data processing, analytics, and visualization to give better insights and ultimately guide better decision-making.



2.4.2 NOSQL DATABASE

Azure Product	Alibaba Cloud Product	Description
Azure Cache for Redis	ApsaraDB for Redis	A fully-managed, in-memory database service that is compatible with Redis protocols. It can process data in key-value pairs with low response latency. It is a popular choice for scenarios where high concurrency and low response latency are necessary.
Azure Cosmos DB	ApsaraDB for MongoDB	A MongoDB-compatible database service that is developed based on the distributed system and high-reliability storage engine of Apsara. It uses a multi-node architecture to ensure high availability and supports elastic scaling, disaster recovery, backup and restoration, and performance optimization.
Azure Time Series Insights	Lindorm	Lindorm is a cloud native multi-model database service. It allows you to store data of all sizes. Lindorm supports low-cost storage and processing of large amounts of data and the pay-as-you-go billing method. It is compatible with the open standards of multiple open source software, such as Apache HBase, Apache Cassandra, Apache Phoenix, OpenTSDB, Apache Solr, and SQL.
Azure Cosmos DB, Azure HDInsight	ApsaraDB for HBase	A cost-effective cloud-based intelligent NoSQL service that is of high scalability and compatible with the open-source HBase. This service provides core benefits such as low storage costs, high throughput and scalability, and intelligent data processing. It also ensures high availability by deploying data centers across regions.

2.4.3 RELATIONAL DATABASE

Azure Product	Alibaba Cloud Product	Description
Azure SQL Database Hyperscale	PolarDB	Cloud-native relational database which provides performance and availability of enterprise databases as well as flexibility and cost-effectiveness of open-source databases. A new generation of "Super MySQL, PostgreSQL or compatible Oracle" supporting up to 100 TB storage.
Azure Database for MySQL	ApsaraDB RDS for MySQL	Availability and performance enhanced MySQL service supporting versions 5.5, 5.6, 5.7, and 8.0.
Azure Database for PostgreSQL	ApsaraDB RDS for PostgreSQL	An on-demand PostgreSQL service with automated monitoring, backup, and disaster recovery capabilities.
Azure SQL Database	ApsaraDB RDS for SQL Server	An on-demand database hosting service for SQL Server with automated monitoring, backup, and disaster recovery capabilities.
Azure Database for MariaDB	ApsaraDB RDS for MariaDB	ApsaraDB for MariaDB is compatible with Oracle and designed with multiple enterprise database features. It uses multiple storage engines, including MySQL InnoDB, to meet different user requirements.

2.4.4 UTILITY & TOOLS

Azure Product	Alibaba Cloud Product	Description
Azure Database Migration Service	Data Transmission Service (DTS)	Alibaba Cloud Data Transmission Service (DTS) helps you migrate data between data stores, such as relational databases, NoSQL databases, and data warehouses. You can use DTS to migrate your data to Alibaba Cloud or between combinations of cloud and on-premises data systems. DTS supports several data replication modes, including data migration, data integration, data synchronization, and change tracking. You can choose a combination of data replication modes that best suit your use cases.
Azure Backup	Database Backup (DBS)	Advanced Database and Application Migration (ADAM) service helps migrate database applications (especially for Oracle) to the cloud. ADAM has been a feature in the DTS (Data Transmission Service).
Data Factory/Data Explorer	Data Management (DMS)	

There are more [database-centric documents](#) for reference.

2.4.5 SERVERLESS

Serverless is provided by most of the database engines of Alibaba Cloud. It is ideal for unpredictable workloads, small – and medium-sized enterprises, or individual developers. Serverless instances allow you to scale CPU and memory resources in real time. Serverless database instances support the pay-as-you-go billing method for computing resources and provide the following benefits: low resource usage, ease of use, flexibility, and low price. The instances can help you quickly and independently scale computing resources to adapt to fluctuating workloads, reduce costs, and improve efficiency.

The following figure shows the changes in resources that are used by a regular instance and a serverless instance if your workloads greatly fluctuate.

- » The serverless instance scales resources based on your business requirements, and a small number of resources are wasted. This improves resource usage.
- » The serverless instances can provide sufficient resources during peak hours, which ensures performance and service stability.
- » You are charged based on the on-demand resources that are used to run your workloads. This significantly reduces costs.
- » No manual specification changes are required. This improves O&M efficiency and reduces costs for O&M administrators and developers.



- » The automatic start and stop feature is supported for the serverless instance. If no connections are established, the serverless instance is automatically suspended to release computing resources. If a connection is established, the serverless instance is automatically started. The process is imperceptible to users.
- » The serverless instance optimizes high-throughput write operations and high-performance batch processing operations and supports auto-scaling. This is suitable for scenarios in which large amounts of data and large traffic fluctuations are involved.

2.5 SECURITY

2.5.1 BUSINESS SECURITY

Azure Product	Alibaba Cloud Product	Description
Content Moderator	Content Moderation	Image and video moderation service that accurately detects inappropriate content.
Azure Sentinel	Fraud Detection	A risk management solution that features real-time analysis and accurate identification, and is suitable for all business scenarios.

2.5.2 CLOUD SECURITY

Azure Product	Alibaba Cloud Product	Description
Azure DDoS Protection	Anti-DDoS	Comprehensive DDoS protection for enterprises to intelligently defend against sophisticated DDoS attacks, reduce business loss risks and mitigate potential security threats.
App Service Certificates	Certificate Management Service (original SSL Certificate)	Certificate Management Service (Original SSL Certificate) allows customers to directly apply, purchase and manage SSL certificates.
Azure Security Center	Security Center	A unified security management system that identifies, analyzes, and notifies you of security threats in real time.
Azure Security Control	Cloud Config	A configuration audit service that tracks and audits configurations of your Alibaba Cloud resources, helping you achieve compliance with resource configurations.
Azure Firewall	Cloud Firewall	Cloud Firewall is one of the first SaaS firewalls deployed on the public cloud. It is your first choice for network security that safeguards your businesses in the cloud.





Azure Product	Alibaba Cloud Product	Description
Azure Bastion	Bastion Host	A unified, efficient, and secure platform that provides cloud-based O&M, access control, and operation audit.
Azure Audit Logs	ActionTrail	A service that monitors and records the actions of your Alibaba Cloud account, including the access to and use of Alibaba Cloud services using the Alibaba Cloud Management Console, calling API operations, or SDKs.

2.5.3 DATA SECURITY

Azure Product	Alibaba Cloud Product	Description
Azure Key Vault	Key Management Service (KMS)	Easily protect, manage, use, and audit your cryptographic keys on Alibaba Cloud
Azure Dedicated HSM	Data Encryption Service	Industry-standard hardware security modules (HSMs) deployed on Alibaba Cloud.

2.5.4 IDENTITY MANAGEMENT

Azure Product	Alibaba Cloud Product	Description
Azure Active Directory	Resource Access Management (RAM)	Secure your cloud resources with Resource Access Management to define fine-grained access permissions for users and groups.
Azure Active Directory	Identify-as-a-Service (IDaaS)	Make identity management a painless experience and eliminate Identity Silos.



2.6 CONTAINER/SERVERLESS/MIDDLEWARE

2.6.1 OBSERVABILITY

Azure Product	Alibaba Cloud Product	Description
Azure Monitor	Application Real-Time Monitoring Service (ARMS)	Build business monitoring capabilities with a real-time response based on front-end monitoring, application monitoring, and custom business monitoring capabilities.
Application Insights	Tracing Analysis	Allows developers to quickly identify root causes and analyze performance bottlenecks for distributed applications.

2.6.2 CONTAINERS

Azure Product	Alibaba Cloud Product	Description
Container Instances	Elastic Container Instance (ECI), Serverless Container Service	An agile and secure serverless container instance service.
Azure Kubernetes Service (AKS)	Container Service for Kubernetes (ACK)	A Kubernetes-based service that ensures high efficiency for enterprises by running containerized applications on the cloud.
Azure Container Registry (ACR)	Container Registry (ACR)	A fully managed cloud-native artifacts management platform. ACR provides end-to-end security, intelligent building, efficient distribution of Docker container images, Helm Charts, and OCI-compatible artifacts.
Open Service Mesh (OSM)	Alibaba Cloud Service Mesh (ASM)	A fully managed service mesh platform that is compatible with Istio. ASM provides consistent traffic management, secure communication with authentication, and service observability in heterogeneous environments.

2.6.3 MESSAGE QUEUES

Azure Product	Alibaba Cloud Product	Description
Service Bus	AlibabaMQ for Apache RocketMQ	AlibabaMQ for Apache RocketMQ is a distributed message queue service that supports reliable message-based asynchronous communication among microservices, distributed systems, and serverless applications.
Service Bus	Message Queue for Apache Kafka	A fully-managed Apache Kafka service to help you quickly build data pipelines for your big data analytics.



Azure Product	Alibaba Cloud Product	Description
Service Bus	Message Queue for RabbitMQ	A distributed, fully managed, and professional messaging service that features high throughput, low latency, and high scalability. This service supports Advanced Message Queuing Protocol (AMQP) and is fully compatible with open-source RabbitMQ.
Service Bus	Message Service	A message queuing and notification service that facilitates the smooth transfer of messages between applications.

2.6.4 MICRO SERVICES

Azure Product	Alibaba Cloud Product	Description
Azure Spring App	Enterprise Distributed Application Service (EDAS)	A PaaS platform for a variety of application deployment options and microservices solutions to help monitor, diagnose, operate, and maintain your applications.
App Configuration	Application Configuration Management (ACM)	Centralizes the management of application configurations and performs real-time configuration push in a distributed environment.

2.6.5 SERVERLESS

Azure Product	Alibaba Cloud Product	Description
Azure Functions	Function Compute (FC)	A secure and stable, elastically scaled, O&M-free, pay-as-you-go, serverless computing platform.
Azure Logic Apps	Serverless Workflow	Visualization, O&M-free orchestration, and Coordination of Stateful Application Scenarios.

2.6.6 APPLICATIONS INTEGRATION

Azure Product	Alibaba Cloud Product	Description
Azure Functions	EventBridge	EventBridge is a serverless event bus service that connects to Alibaba Cloud services, custom applications, and SaaS applications as a centralized hub. EventBridge can also use the CloudEvents 1.0 specification to route events among these services and applications. EventBridge helps you build loosely coupled and distributed event-driven architectures.





3. MIGRATION PROCESS

Alibaba Cloud offers a full range of services and resources to help you comprehensively assess and plan your migration journey, quickly migrate to Alibaba Cloud in easy steps with enhanced security measures, and manage your resources on Alibaba Cloud with cost-effective services to ensure business continuity. There are 4 major steps:



3.1 MIGRATION ASSESSMENT

Evaluate Migration Needs

Evaluate the challenges and objectives for migration, understand the cloud environment and the tolerance to maximum downtime, current or potential risks of failure, and the compatibility of the current system to the target cloud platform.

Estimate TCO of the Cloud Environment

Calculate the Total Cost of Ownership (TCO) of cloud infrastructure and compare it with the cost of other cloud vendors using Cloud Migration Hub.

Prepare the Resource Inventory List

Analyze your current resource inventory to forecast cloud infrastructure requirements and migration workloads to gain more visibility on migration time, method, and scheduling.



3.2 MIGRATION PREPARATION

Design Migration Strategy

Gain a full understanding of Alibaba Cloud's solutions for migrating your IT infrastructure to Alibaba Cloud, including application migration, data migration, and database migration-related tools, utilities, and methodologies, to plan and implement an IT infrastructure migration.

Map Products

Blueprint the overall architecture and map the appropriate cloud products and services based on your business needs to ensure high availability, scalability, and performance.

Prepare for Cloud Governance

Pre-plan the IT resources of different accounts, including landing zone deployment, resource structures, identity and permission management, and security and regulatory compliance, to facilitate a smooth and cost-efficient migration.

Test the Cutover Solution

Verify the cutover strategies to ensure the feasibility of the cutover solution, such as validating applications to ensure the cutover process is in order, identifying gaps and issues, and designing cloud backup strategies for cutover applications.

3.3 MIGRATION IMPLEMENTATION

Migrate Application Servers

You can use Alibaba Cloud [Server Migration Center \(SMC\)](#) to migrate one or more source servers to Alibaba Cloud. Source servers can be servers in data centers, VMs, servers on other cloud platforms, and servers of other types. For agentless migration of VMWare virtual machines, you can use Alibaba Cloud [Hybrid Backup Recovery \(HBR\)](#) to ensure cost-effective disaster recovery and migration for data stored in VMware virtual machines.

Migrate Containers

You can migrate containers to Alibaba Cloud with your own self-built platform or the tools we provide. First, create and configure resources on the target clusters, then migrate data to corresponding Alibaba Cloud services. For example, migrate images to [Container Registry](#), databases to [ApsaraDB RDS for MySQL](#) and [PolarDB](#), and data storage to [Object Storage Service](#) and [File Storage NAS](#). Next, you can choose to migrate applications with your self-built CI/CD platform, or with Alibaba Cloud [Velero](#). Last, test the services of your



business after migration, and update the DNS configuration to switch traffic to ACK clusters. You can also migrate docker image files with Alibaba Cloud [Image Syncer](#) to Alibaba Cloud Container Registry.

Migrate Databases

Alibaba Cloud [Database Migration](#) solution provides a low-cost self-service database migration experience. Both homogenous migration such as MySQL to MySQL, and heterogeneous migration such as Oracle Database to PolarDB, are supported. You can smoothly migrate databases from 100s of GB to multiple TBs in size with minimal business impact. You can also leverage the [Oracle Database Migration](#) solution to migrate your legacy Oracle databases to Alibaba Cloud – to save on long-term costs and take advantage of improved scalability, reliability, robust security, high performance, and cloud-native features.

Migrate Data

You can migrate data from a local IDC to Alibaba Cloud OSS, or from an on-premises Network Attached Storage (NAS) server to Apsara File Storage NAS or OSS for long-term storage. You can also migrate the data of a web-based service provider from a cloud service to Alibaba Cloud Object Storage Service (OSS). Please refer to [Data Online Migration](#) for detailed migration instructions based on your business needs.

3.4 OPERATION AND MONITORING

Monitor Cloud Resources

Monitor the usage of the cloud resources, and the status and health of your business to ensure the availability of your application when an alarm is triggered.

Manage Operations

Manage O&M permissions in a centralized manner, monitor all O&M operations, and reproduce O&M scenarios in real-time to facilitate identity authentication, access control, and operation audit.

Ensure Business Continuity

Improve the business availability on the cloud with secure, efficient, and cost-effective backup services, cross-AZ and cross-region disaster recovery, deduplication and AES-256 encryption features.

Automate Infrastructure Deployment

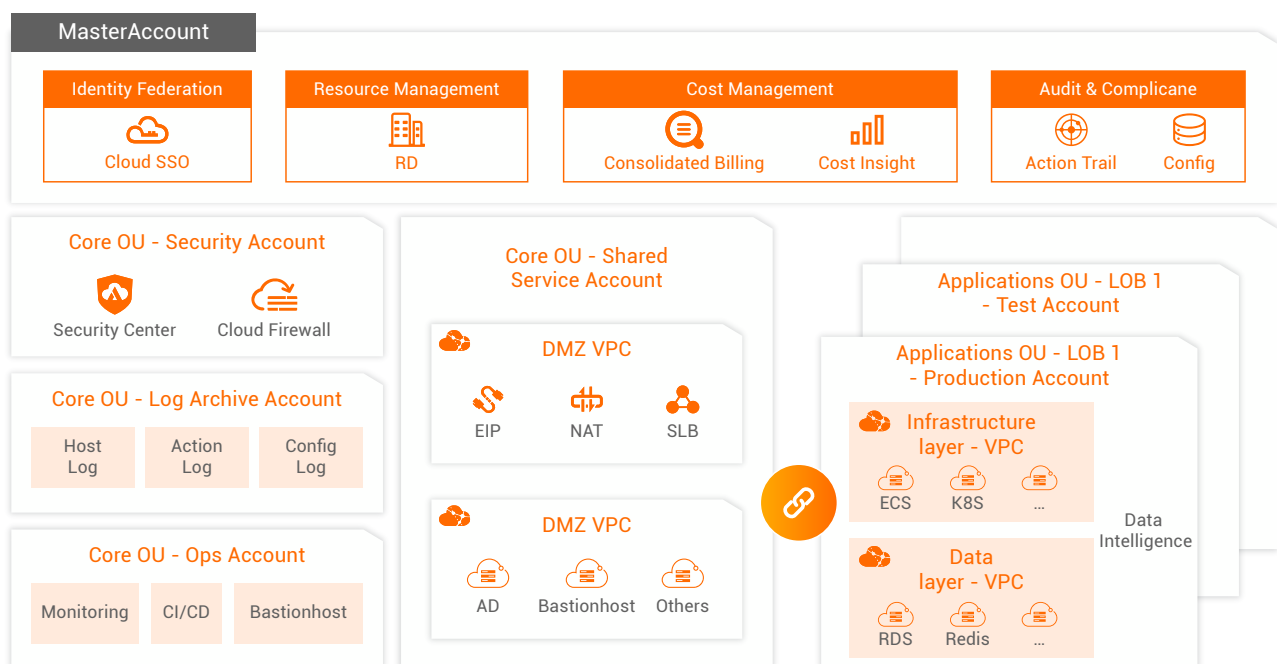
Manage identical future infrastructure deployment with Infrastructure as Code (IaC) to reduce errors that will compromise the cloud infrastructure and application deployment.



4. MIGRATION PRACTICES

4.1 CONSIDER YOUR ACCOUNT STRUCTURE AND IAM – LANDING ZONE

In order to ensure the operational safety, compliance, manageability, and scalability of the cloud business, while accelerating the speed of cloud migration, we recommend that customers design and build Alibaba Cloud Landing Zone before the migration. When migrating, priority is given to the two neighborhoods of resource planning, and identity authority. The typical architecture design is as follows.



To help you better understand the basic products of Alibaba Cloud, the following describes the relationship between each cloud product to build its core parts. In the actual deployment process, Alibaba Cloud has provided a [Cloud Governance Center](#) that contains common Landing Zone blueprints to help you accelerate deployment.



4.1.1 MIGRATE AZURE SUBSCRIPTION

Azure Subscription is the basis of LandingZone. A subscription in azure usually corresponds to an Alibaba Cloud account. Azure subscriptions have hierarchical relationships, and Alibaba Cloud provides a multi-account architecture to correspond to this hierarchical relationship. The following will explain how to set up a Landing Zone multi-account structure in Alibaba Cloud.

4.1.1.1 DETERMINING THE MASTER ACCOUNT

The first step is to confirm the master account of the enterprise. The enterprise management master account includes the [Resource Directory](#) (RD), which is used to manage multiple accounts. You can use this account as a financial custody account, use it to make unified payments and settlements for other accounts, and view cloud bills in a unified manner. Unify the audit rules configured by this account and send them to each member's account.

4.1.1.2 CREATING RESOURCE DIRECTORY

Master account Resource Directory (RD) is enabled, and a root folder is automatically created corresponding to the Azure Tenant Root Group.

Azure Management groups	Alibaba Cloud Resource Directory
Tenant Root Group	Root Folder

You can create a folder under it. We recommend that you create at least two folders, a core folder, and an applications folder, to distinguish your management account from your business account. If your enterprise has more complex management levels and more businesses, you can also create more business folders at multiple levels.

- » Core: Centrally manages public service accounts
- » Custom: Centrally manages service accounts

4.1.1.3 CREATING A PUBLIC SERVICE ACCOUNT

After a folder is created, the resource directory provides an easy account creation function. Accounts created through the resource directory automatically inherit the real-name authentication of your master account enterprise. We recommend that you create at least three public service accounts for different management scenarios.



1. Shared service account

Corporate shared services will be deployed in this account, such as network configuration, unified access to the public network configuration, etc.

For network-related cloud products, refer to the network section.

2. Log account

Which aggregates the logs of all member accounts. Including cloud product operation behavior logs, audit logs, and business logs.

Azure	Alibaba Cloud
Activity Log	Action Trail
Policy Resource Graph	Cloud Config

3. Security account

Generally, the security role in the enterprise will use this account to configure related security products. Such as common Web Application Firewall (WAF), high-defense IP, etc.

For security-related cloud products, refer to the security section.

4.1.2 ACCESS CONTROL

In Alibaba Cloud, we use products and services such as access control (RAM) to achieve identity management and access control capabilities. This part is equivalent to Azure's policy function.

RAM is an advanced access control mechanism based on attribute-based access control (Attribute-Based Access Control, ABAC). When a customer request arrives at Alibaba Cloud, Alibaba Cloud will evaluate the currently accessed request characteristics, identity characteristics, and resource characteristics, and match them with the configured permissions of the identity to complete the authentication.

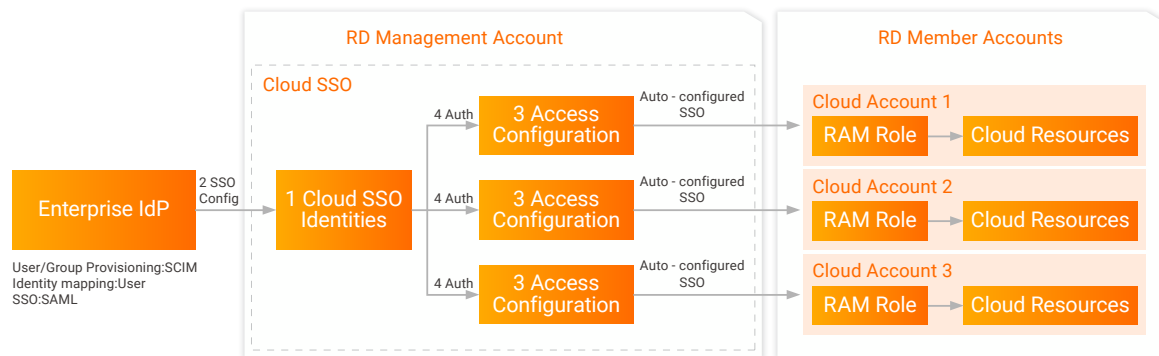
The identity and access control granularity supported by each cloud product can be found in cloud services that [support RAM](#) and cloud services that [support STS](#).

4.1.3 SINGLE SIGN-ON (SSO)

In the multi-account scenario, each account needs to manage its identity permissions and configure the single sign-on between each account and Alibaba Cloud. This management cost is very high.



We recommend using Alibaba Cloud's [CloudSSO](#), which can integrate the multi-account unified identity management and access control of the Alibaba Cloud Resource Directory (RD).

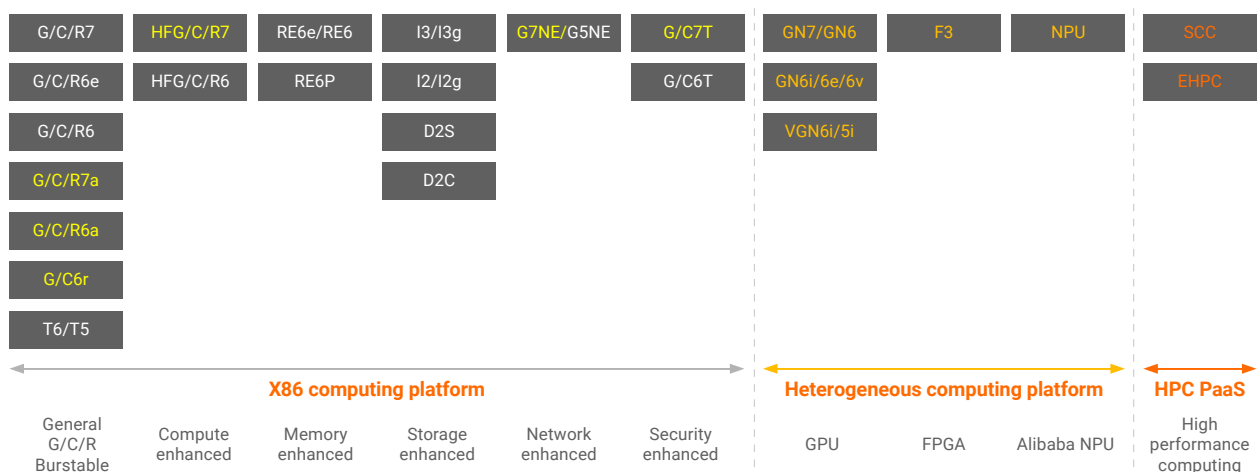


4.2 COMPUTE SERVICE

4.2.1 CHOOSE ECS INSTANCE TYPE ACCORDING TO AZURE VM

Before migrating Azure VM to Alibaba cloud, you have to choose the corresponding instance type carefully. With appropriate instance types, it can improve the application performance while reducing the total cost of computing resources.

Here is the diagram to show the Alibaba Cloud instance families portfolio. You can find what types of workloads or scenarios running on each instance family on the Alibaba Cloud [website](#).



Here are the Alibaba ECS instance types compared with the Azure VM types side by side, which is a brief guideline on how to select the appropriate Alibaba Cloud instance families.

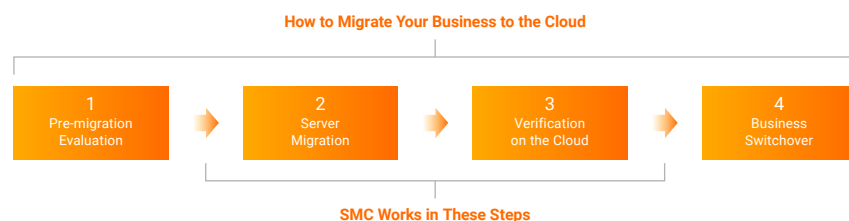
Product Category	Azure	Alibaba Cloud
Burstable	B-series VM	T6/T5
General Purpose (Intel CPU)	Dv5/Dsv5/Ddsv5/Dv4/Dsv4/Ddsv4/Dv3/Dsv3	G7/G6e/G6
Compute Optimized (Intel CPU)	Fsv2	C7/C6e/C6
Memory Optimized (Intel CPU)	Ev5/Esv5/Edsv5/Ev4/Esv4/Edsv4/Ev3/Esv3	R7/R6e/R6
Memory Enhanced	M series	RE6/RE4
Security Enhanced	Dcsv2 (SGX)	C7t/G7t/R7t (SGX)
High Frequency	c2-standard	HFC7/HFR7/HFG7/HFC6/HFR6/HFG6
Instance with Local SSD	N/A	I3/I3g/I2ne/I2gne
Instance with Local HDD	N/A	D2S/D2C
GPU Instances	NCA100v4/NDasrA100v4/NdmsA100v4/NvadsA10/Ncv3/NcasT4v3	Gn7e/Gn7i/Gn6e/Gn6i
AMD Instances	Da series/Ea series / HB series	C7a/G7a/R7a
ARM Instances	Epsv5/Epdsv5-series	C6r/G6r

4.2.2 USING SMC TO MIGRATE VM FROM AZURE TO ALIBABA CLOUD

4.2.2.1 WHAT IS SMC?

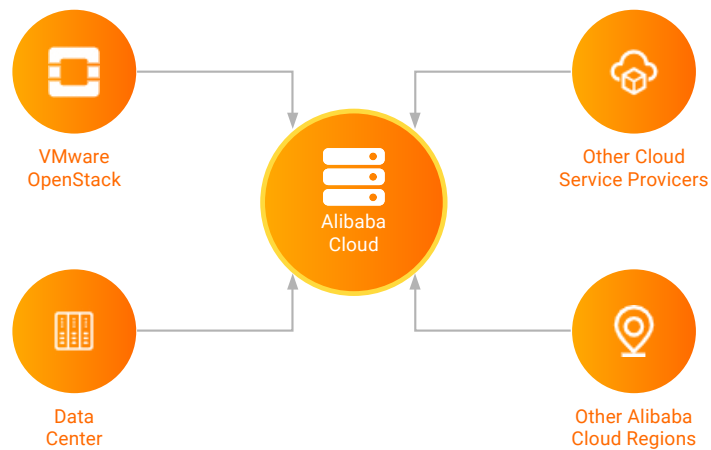
Alibaba Cloud Server Migration Center (SMC), is a free tool for Cloud Migration launched in November 2017. SMC allows you to migrate one or more source servers to Alibaba Cloud. Source servers can be servers in data centers, virtual machines, servers on other cloud platforms, and other types of servers. SMC helps you automate the migration of server application environments and simplifies the migration of system images to ECS instances. To facilitate the management of migration tasks, SMC provides service-oriented API operations that allow you to create and execute migration scripts and query the migration progress.

4.2.2.2 HOW SMC CAN HELP?



SMC works in server migration and can help to verify the VM after migration. Before SMC migrates VM, application pre-migration evaluation should be done. We suggest migrating the VM using the pay-as-you-go purchasing model before the business shift to Alibaba Cloud. After that, you can change PAYG to other purchasing models such as monthly/yearly subscriptions for those all-time running workloads.

Note For database migration, DTS is recommended.



SMC allows you to migrate servers from your own data centers, your local virtual machines (VMware, Virtual Box, Xen, and KVM), and other cloud platforms such as AWS, Azure, Google Cloud, Huawei Cloud, Tencent Cloud, UCloud, Telecom Cloud, and QingCloud.

4.2.2.3 KEY BENEFITS OF SMC



Zero Dependency on Underlying Environment

SMC supports both P2V and V2V, multiple file system formats, and multiple disk types.



Zero Local Storage

Data is directly migrated to the cloud and does not consume local storage.



Zero Downtime

The migration does not affect the services of the source system and does not cause downtime.



Lightweight and Portable

The SMC client is small and does not require installation. It involves simple deployment.



Flexible Configuration

SMC provides a variety of migration and transmission solutions that you can choose from.

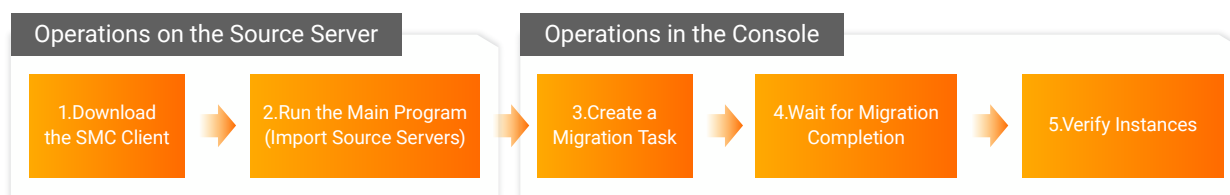


One-click Migration

The migration is managed. You can start migration with a few clicks.



4.2.2.4 SMC MIGRATION STEPS



You have to make sure the following things are well prepared before you perform SMC migration:

- » [RAM and AccessKey are prepared](#)
- » Network connection (either public or private network) established between source VM and destination VM.
- » Destination network environments created including VPC, vSwitch, and security group.

There are essentially 5 steps to migrate a VM to Alibaba Cloud:

- 1.** Download and install the SMC client on your source server.
 - » The client has 32-bit and 64-bit versions for Windows and Linux, please select the right one
 - » `go2aliyun_client` is the main program
 - » `user_config.json` is the user configuration file.
- 2.** Run the main program in the source server to import the source server information to Alibaba Cloud SMC service.
- 3.** Create a migration task in the Alibaba Cloud SMC console and start the job.
- 4.** Migration will run automatically until completion; the time spent depends on the source server data capacity and outbound bandwidth.
- 5.** Verify the migration result.

Here are some useful links for SMC tools and, if you have any trouble using this tool, you can contact the Alibaba Cloud GTS (Global Translation Service) team for support:

- » [SMC Best Practice](#)
- » [SMC Console](#)
- » [SMC Help](#)



4.3 NETWORKING

4.3.1 SLB

4.3.1.1 CLB DEPLOYMENT

CLB is a product of Alibaba Cloud's load-balancing product line. It provides layer 4 and basic layer 7 load-balancing forwarding capabilities, corresponding to Azure LB products. The creation and use of CLB is very simple. You can use the console to create and manage CLB instances, or you can use APIs to operate. Please refer to [help documents for console operations](#).

4.3.1.2 DIFFERENCES WITH AZURE LOAD BALANCER

Azure doesn't have the counterpart of Alibaba Cloud CLB, as it only has the counterpart of NLB and ALB, which are Load Balancer and Application Gateway, and works on layer 4 and layer 7 only. While Alibaba Cloud CLB could work on both layer 4 and layer 7, we recommend using the NLB and ALB instead.

4.3.1.3 ALB DEPLOYMENT

ALB is one of the products of Alibaba Cloud's load-balancing product line. It is a product specifically oriented to layer 7 application load balancing. It provides many powerful functions based on layer 7 services, both in terms of internet and ALB computing power. With the ability to automatically scale up and down, it can greatly simplify the configuration, deployment, and planning of customers, and focus on application deployment. You can use the ALB console or API to create and manage ALB instances. Please refer to the [console operation help documents](#).

4.3.1.4 NLB DEPLOYMENT

[NLB](#) is the latest product of Alibaba Cloud's load-balancing product line.

4.3.2 VPC

4.3.2.1 VPC DEPLOYMENT

Alibaba Cloud VPC is the counterpart of Azure Virtual Network. It is suggested to create an Alibaba Cloud VPC that has a different CIDR block from your current Azure VNet to avoid conflict. This will definitely help the resource migration because both cloud platforms don't support layer 2 connections on physical links. Please refer to the [console operation help documents](#).



4.3.3 NAT GATEWAY

4.3.3.1 NAT GATEWAY DEPLOYMENT

NAT Gateway is one of the basic network element products of Alibaba Cloud. It provides the function of IP NAT for the public network and the internal network, which can meet the mainstream network function requirements of customers. You can use the [help documents to create and manage NATGW instances](#).

4.3.3.2 DIFFERENCES WITH AZURE NAT

- » Azure NAT doesn't support DNAT, since DNAT is integrated into their LB product. Customers using DNAT on Azure LB could fulfill it on Alibaba Cloud NAT Gateway product.
- » Azure NAT supports 50Gbps throughput by default, while Alibaba Cloud supports 15Gbps. If there is a higher requirement, the customer can contact the product team to adjust.
- » Azure NAT supports TCP timeout adjustment, while Alibaba Cloud doesn't support it for now.

4.3.4 BUILD A PRIVATE CONNECTION BETWEEN ALIBABA CLOUD AND MICROSOFT AZURE

4.3.4.1 EXPRESS CONNECT

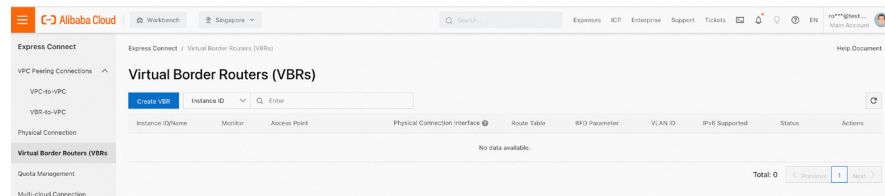
Whether it is Alibaba Cloud or Azure, there are many dedicated leased line partners who can provide dedicated interconnection between Alibaba Cloud and Azure, like China Unicom/China Mobile/China Telecom/Equinix, and more. Customers can purchase multi-cloud connection products through third-party service providers for the dedicated line interconnection between Alibaba Cloud and Azure. Physical links are generally divided into the exclusive type and shared type. Lines of dedicated type require service providers to perform fiber construction through LOA (usually in-building lines). The cycle is relatively long, and two specifications of 1G/10G ports are provided; Lines of shared type are opened pre-built dedicated lines by ISP and can provide various specifications from 50Mbps to 1Gbps, the cycle is relatively short compared to dedicated type.



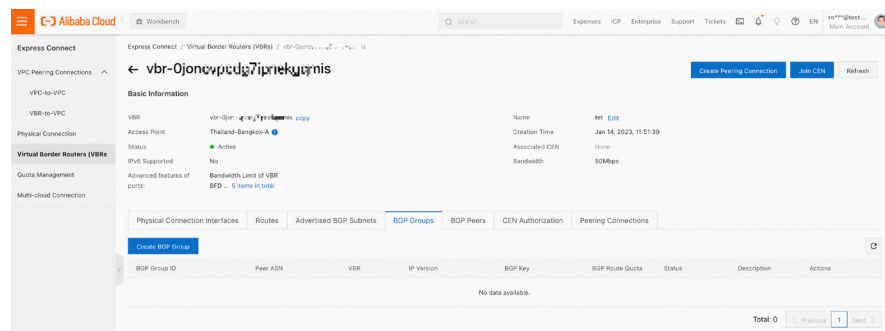
Configurations:

Alibaba Cloud side

1. After the creation of the Network Edge is complete, a VBR instance will appear on the Aliyun console Express Connect – Virtual Border Router. Note that the Singapore region is selected in the upper left corner:

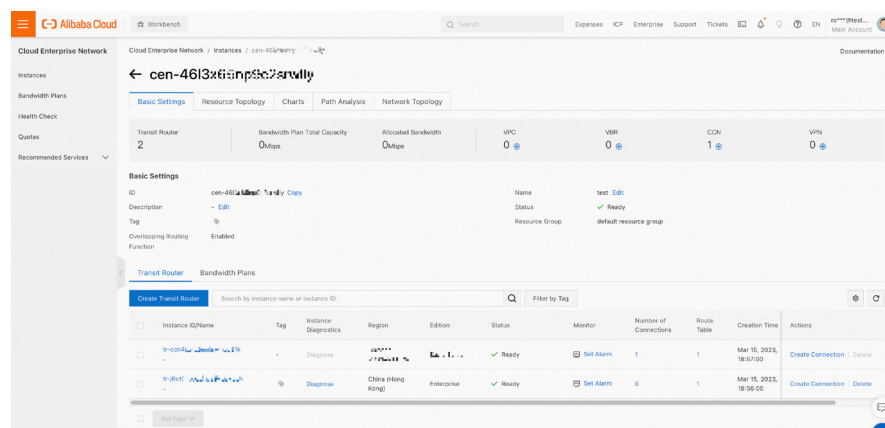


2. Click on the VBR instance to create a BGP group and BGP neighbor:

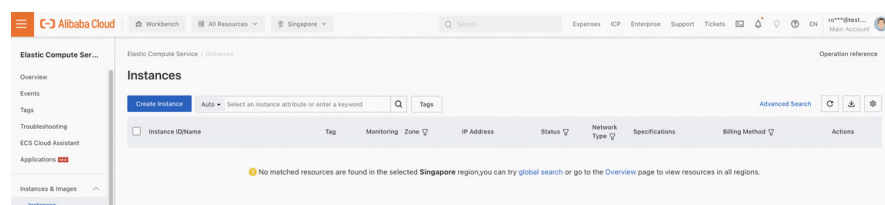


After the configuration is complete, you can see that the BGP negotiation status is Established:

3. Create a VPC and a Cloud Enterprise Network(CEN) in the Singapore region, and load the VPC and VBR into the CEN:



4. Create two ECSs in the VPC to simulate two different types of users:



Azure side

You can refer to [this link](#) to build a private link between Alibaba Cloud and Azure.

4.3.4.2 VPN GATEWAY

VPN Gateway is an internet-based service that securely and reliably connects enterprise data centers, office networks, and third-party public clouds like AWS or Azure to virtual private clouds (VPCs) of Alibaba Cloud through encrypted channels.

Configurations:

- » [Alibaba Cloud side](#)
- » [Azure side](#)

4.3.4.3 SMART ACCESS GATEWAY VCPE

SAG vCPE provides an image that can be deployed on data center servers, Edge Node Service (ENS) instances, Alibaba Cloud instances, Amazon Web Services (AWS) instances, and Microsoft Azure instances. After you deploy the SAG vCPE image on an instance, the instance serves as a vCPE device. The bandwidth of private networks for encrypted connections can reach 300 Mbit/s and higher (the packet length in the performance test is 1,024 bytes). This allows you to connect private networks to Alibaba Cloud in a more flexible way.

Configurations:

<https://www.alibabacloud.com/help/en/smart-access-gateway/latest/connect-microsoft-azure-resources-to-alibaba-cloud-resources-through-sag-vcpe>

4.3.4.4 HIGH AVAILABILITY

Two physical leased lines

Two physical leased lines can work in an Active/Active or Active/Standby manner. Both Alibaba Cloud and Azure support the feature.

- » [Alibaba Cloud](#)
- » [Azure](#)

Physical leased line with VPN Gateway

You can configure the function of automatically switching to IPsec VPN when the physical leased line fails.

- » [Alibaba Cloud](#)
- » [Azure](#)





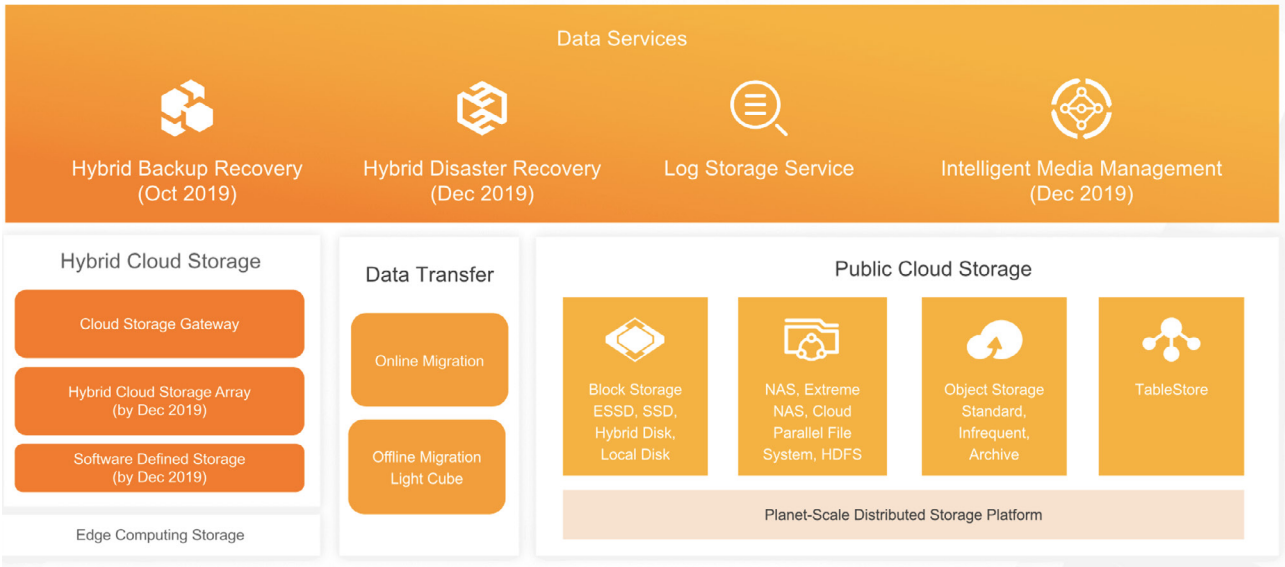
Multiple VPN tunnels

You can configure multiple IPsec VPN tunnels to realize the function of mutual backup.

- » [Alibaba Cloud](#)
- » [Azure](#)

4.4 STORAGE

For storage, Alibaba Cloud provides many products and services. The first step to consider is data migration for the foundational storage products.



4.4.1 DATA MIGRATION

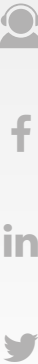
When migrating from Azure, your data might come from the following sources:

Source	Destination
Azure Blob	OSS
Azure Files	NAS

When you migrate your data from Azure Blob to Alibaba Cloud or perform any cross-cloud data migration in general, you need to consider the following things:

Data size and timeline

If the total data size is within a few TBs and the migration task is not time-sensitive, consider using Online Migration through the internet. If the total data size is 100TB and above, consider using Online Migration through the express route. For example, the table below shows sample figures to see how long a migration job can take for reference. In real scenarios, the time to migrate also depends on the actual bandwidth and number of items to be migrated.



Data to migrate	Bandwidth	Time to migrate
1TB	1Gb/s (Internet)	140 minutes
1TB	10Gb/s (Internet)	14 minutes
1TB	100Gb/s (Express route)	1.4 minutes
1PB	1Gb/s (Internet)	100 days
1PB	10Gb/s (Internet)	10 days
1PB	100Gb/s (Express route)	1 day

Cost

The cost for data migration includes the following:

Fees	Description
Internet Egress fee from Azure Blob	If you choose approach 1, Azure charges the internet egress fee.
Direct Connect Egress fee from Azure Blob	If you choose approach 2, Azure charges the egress fee for Direct Connect.
API request fee from Azure Blob	Azure charges API requests that will be used to retrieve the data from Blob.

For example, the egress fee to the internet is as below. Please refer to this link [here](#) for more details.

Data Transfer OUT From Amazon S3 To Internet

Up to 1 GB / Month	\$0.00 per GB
Next 9.999 TB / Month	\$0.09 per GB
Next 40 TB / Month	\$0.085 per GB
Next 100 TB / Month	\$0.07 per GB
Greater than 150 TB / Month	\$0.05 per GB

Cross-Region migration

If you plan to migrate to another region, it is recommended to use a close-by region as an intermediate region and perform cross-region replication for the initial migration.

4.4.1.1 MIGRATION DATA FROM AZURE BLOB TO OSS

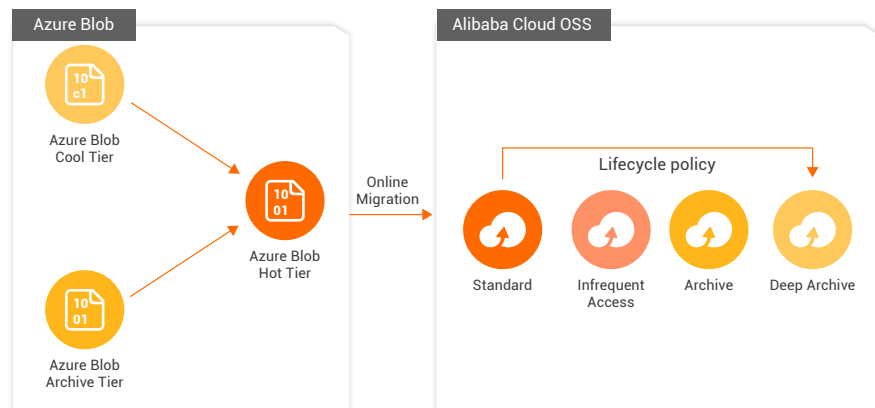
Choose the right storage class

Below is a mapping between different storage classes of Azure Blob and OSS. Choose the right storage class.

Azure Blob Class	OSS Cloud Storage Class
Online – Hot	Standard
Online – Cool	Infrequent Access
Archive	Archive
Archive	Cold Archive



Since the Hot tier is optimized for frequent read and write, data migration should be done using the Hot tier to optimize for the migration cost.



Migration from Azure Blob Hot Tier to OSS via Internet

The online migration feature in [Data Transport](#) is a serverless migration engine that natively allows you to migrate your Azure Blob Hot Tier buckets to OSS. Use this tool only when you perform Online Migration via the internet. The tool allows you to create Azure Blob and OSS buckets as target addresses and then create migration jobs between the two targets.

Migration from Azure Blob Hot Tier to OSS via Express Connect

[ossimport](#) is a tool used to migrate data to OSS. You can deploy ossimport on local servers or ECS instances in the cloud to migrate data stored locally or in other cloud storage systems to OSS. It supports standalone mode (for <30TB migration) and distributed mode for (>=30TB migration).

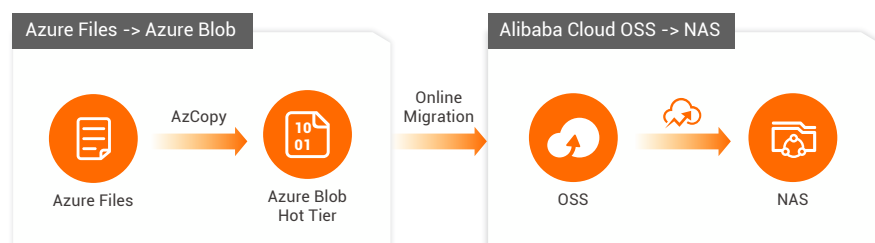
Migration from Azure Blob Cool or Archive Tier to OSS

If you plan to migrate from Cool Tier or Archive Tier to OSS Archive or OSS Cold Archive, it is recommended to convert to the Hot Tier and then follow the Hot Tier to OSS approach for data migration, to avoid significant data retrieval costs from the Cool or Archive Tier.

4.4.1.2 MIGRATION FROM AZURE FILES TO NAS

Migration from Azure Files to NAS leverages Azure Blob to OSS as an interim migration staging area.

- » [Use Azcopy to transfer data from Azure Files to Azure Blob](#)
- » Migrate Azure Blob to OSS as described in the previous section
- » Use Online Migration to [transfer OSS to NAS](#)



4.4.2 APPLICATION MIGRATION

For object storage, it is recommended that you switch from Azure Blob SDK to OSS SDK. While most of the APIs are compatible, there are a few differences. Refer to this document for [more details](#).

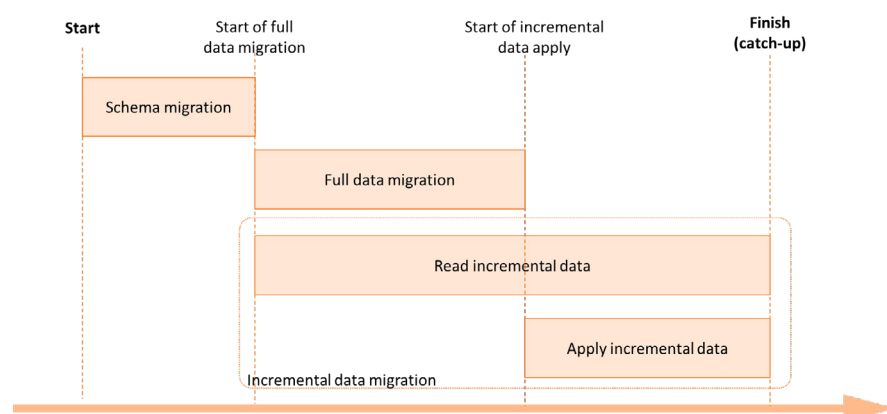
During the transition, your previous application might be still generating data and writing to Azure Blob. And your new application based on OSS needs to be able to read data that is newly generated. The back-to-origin setting allows you to retrieve the data from Azure Blob when the data is not available in OSS. Follow this document for [more details](#).

4.5 DATABASE

Database migration is always a cornerstone for cloud migration either from on-premise to the public cloud or from one cloud platform to another cloud platform. The database scheme is pretty clear and also with sophisticated migration techniques due to data reliability, high availability, application compatibility, and business continuity. Please also refer to the [overview of the database migration to Alibaba Cloud](#).

Specifically, we cover most of the cases when migrating databases from Azure to Alibaba Cloud. We list the migration guide according to the types of databases below. Even for the same type of database, there may be more than one kind of alternative database cloud service as the target database on Alibaba Cloud.

Before heading into the database migration guide, let's talk more about [DTS](#), which is the key cloud service working as the data pipeline between the source and target databases, and works for [major migration scenarios](#). There are commonly three task types in DTS for migration:



Migration type	Description
Schema migration	DTS migrates the schemas of the required objects from the source database to the destination database. Tables, views, triggers, and stored procedures can be migrated. For schema migration between heterogeneous databases, DTS converts the schema syntax based on the syntax of the source and destination databases. For example, it converts the <code>NUMBER</code> data type in Oracle databases into the <code>DECIMAL</code> data type in MySQL databases.
Full data migration	DTS migrates historical data of the required objects from the source database to the destination database. If you select only Schema Migration and Full Data Migration as the migration types, DTS does not migrate incremental data generated in the source database. Note To ensure data consistency, we recommend that you do not write data to the source database during full data migration. To ensure service continuity, you must select Schema Migration, Full Data Migration, and Incremental Data Migration when you configure a data migration task.
Incremental data migration	DTS retrieves static snapshots that are used for full data migration from the source database and migrates the snapshot data to the destination database. Then, DTS synchronizes incremental data generated in the source database to the destination database. Note During incremental data migration, data is synchronized between the source and destination databases in real-time. The migration task does not automatically stop. You must manually stop the migration task.

4.5.1 GENERAL STEPS

Usually, there are applications running on top of the target database system on Azure. Before we take the actual migration actions, here are the general steps for most cases.

1. Target database planning and selection. Following the questions to determine the target database on Alibaba Cloud.

Which database to use on Alibaba Cloud?

- » Cloud-hosted Alibaba Cloud Database vs Self-built database on ECS. Please refer to [ApsaraDB for RDS vs. Self-built Databases](#) for a brief comparison. Essentially, we recommend the [cloud-hosted Alibaba Cloud Database](#) for improved reliability, serviceability, cost-efficiency, upstream and downstream cloud service integration etc.
- » Alibaba Cloud Database Options. In case there are multiple options for a single type of database migration (e.g., RDS MySQL or PolarDB MySQL in section 2.4.1 MySQL migration, PolarDB Oracle, RDS MySQL, PolarDB MySQL, or AnalyticDB PostgreSQL in section 2.4.4 Oracle migration), we will elaborate the key difference in each section.



Which region, edition, and specification to use?

- » Please refer to the product purchase pages for database product availability in global regions.

Database Service	Global Region & Purchase Page
ApsaraDB RDS	Global Region & Purchase Page
PolarDB	Global Region & Purchase Page
Redis	Global Region & Purchase Page
MongoDB	Global Region & Purchase Page
HBase	Global Region & Purchase Page
TSDB (Time Series Database)	Global Region & Purchase Page
InfluxDB	Global Region & Purchase Page
AnalyticDB MySQL	Global Region & Purchase Page
AnalyticDB PostgreSQL	Global Region & Purchase Page
ClickHouse	Global Region & Purchase Page

2. Network connectivity setup

- » DTS is the key service for the actual migration procedure. The network connectivity is to make sure the DTS could connect to the target database on Azure. DTS itself is a cloud service on Alibaba Cloud, after the target database has been determined and planned, it is recommended to create a DTS migration task at the same region with the target database on Alibaba Cloud (please refer to [this link](#) for creating DTS migration task in global regions).
- » Then please make sure the network connectivity from Alibaba Cloud to the source Azure is working either through the public IP: Port for database connection from DTS to the source database, or the network is connected via Cloud Enterprise Network(CEN), Express Connect, VPN Gateway, or Smart Access Gateway.
- » Then please refer to [this link](#) to add the CIDR blocks of DTS servers at a specific region, where the DTS migration task is located, to the security settings of the source Azure database.



3. Database migration

» When the prerequisites are ready, please walk through the following sections for the actual migration procedure. Usually, the user cares about the connection status and data transmission performance during the DTS full data migration and incremental data migration. Please refer to the following documents:

- [View the connection status and performance of full data migration](#)
- [View the connection status and performance of incremental data migration](#)

4. Application switch over

» When the target database is running, the database connection Uniform Resource Identifier (URI) needs to be updated in the application to switch over to the new database on Alibaba Cloud.

4.5.2 MYSQL

Source database on Azure	Target database on Alibaba Cloud	Migration guide
Self-built MySQL on Virtual Machine	ApsaraDB RDS for MySQL This is the most common migration scenario with 100% MySQL compatibility.	» Migration guide
	PolarDB MySQL Migrate to cloud-native MySQL with computing and storage decoupled architecture to gain fast elasticity and up to 100TB data storage.	» Migration guide
Azure Database for MySQL	ApsaraDB RDS for MySQL This is the most common migration scenario.	» Migration guide
(There is no specific guide for migrating Azure Database for MySQL, please refer to the general guide of self-managed MySQL migration.)	ApsaraDB RDS for MySQL When a user wants to migrate to the most commonly used RDS MySQL.	» Migration guide
	PolarDB MySQL This is the most common migration scenario.	» Migration guide



4.5.3 POSTGRESQL

Source database on Azure	Target database on Alibaba Cloud	Migration guide
Self-built PostgreSQL on Virtual Machine	ApsaraDB RDS for PostgreSQL	» Full data migration » Incremental data migration for PostgreSQL versions 9.4.8 to 10.0 » Incremental data migration for PostgreSQL versions 10.1 to 13
Azure Database for PostgreSQL (There is no specific guide for migrating Azure Database for PostgreSQL, please refer to the general guide of self-managed PostgreSQL migration.)	ApsaraDB RDS for PostgreSQL	» Full data migration » Incremental data migration for PostgreSQL versions 9.4.8 to 10.0 » Incremental data migration for PostgreSQL versions 10.1 to 13

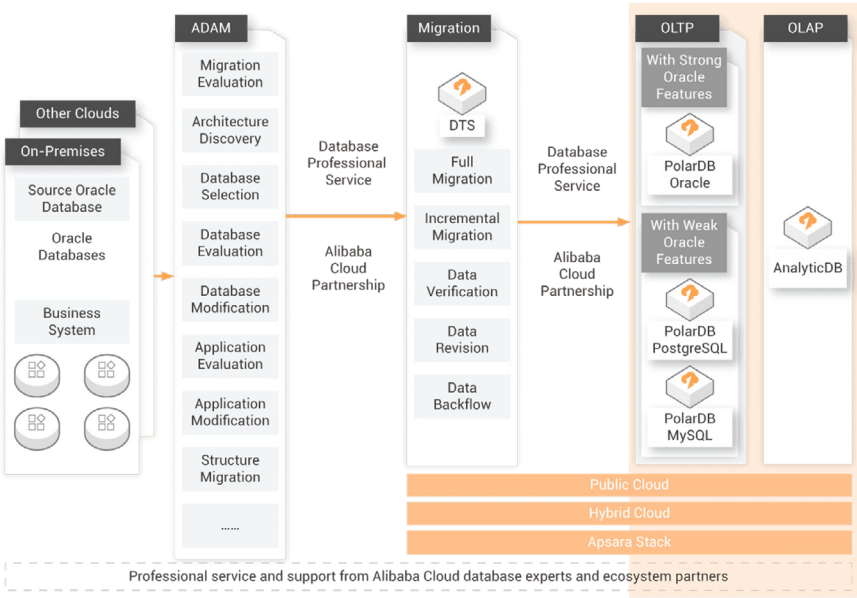
4.5.4 SQL SERVER

Source database on Azure	Target database on Alibaba Cloud	Migration guide
Self-built SQL Server on EC2	ApsaraDB RDS for SQL Server	» Full data migration » Incremental data migration
Amazon RDS for SQL Server	ApsaraDB RDS for SQL Server	» Full data migration



4.5.5 ORACLE

It would take more time for migrating Oracle from Azure to cloud-native databases on Alibaba Cloud. We provide a comprehensive [Oracle Migration](#) on Alibaba Cloud.



Source database on Azure	Target database on Alibaba Cloud	Migration guide
Self-built Oracle on Virtual Machine	PolarDB Oracle We recommend this migration option with cloud-native features, high compatibility with Oracle, but with cost reduction.	» Migration guide
	ApsaraDB RDS for MySQL If you decide to migrate to the MySQL ecosystem with strategic consideration in the long term, this can avoid vendor lock-in.	» Migration guide
	PolarDB MySQL If you decide to come to MySQL ecosystem (with cloud-native MySQL on Alibaba Cloud) with strategic consideration in the long term, this can avoid vendor lock-in.	» Migration guide
	AnalyticDB PostgreSQL Migrate from Oracle RAC (Real Application Clusters) or Oracle data warehouse in most cases.	» Data migration » Application migration
	Message Queue Instance: Alibaba Cloud Kafka Connect to Oracle on Amazon and get full data and incremental data to the message queue. The downstream application or data system then consumes message data from Kafka.	» Migration guide



4.5.6 IBM DB2

Source database on Azure	Target database on Alibaba Cloud	Migration guide
Self-built IBM Db2 on EC2	ApsaraDB RDS for MySQL If you decide to come to the MySQL ecosystem with strategic consideration in the long term, this can avoid vendor lock-in.	» Migration guide

4.5.7 AZURE COSMOS DB FOR MONGODB

Source database on Azure	Target database on Alibaba Cloud	Migration guide
Azure Cosmos DB for MongoDB	ApsaraDB for MongoDB	» Migrate data from Azure Cosmos DB for Mongo to ApsaraDB for MongoDB by using mongodump and mongorestore

4.5.8 REDIS

Source database on Azure	Target database on Alibaba Cloud	Migration guide
Azure Cache for Redis (There is no specific guide for migrating Azure Cache for Redis, please refer to the general guide of self-managed Redis migration.)	ApsaraDB for Redis	» Migrate an Amazon DynamoDB database to ApsaraDB for MongoDB by using NimoShake » Migrate data from Amazon DynamoDB to ApsaraDB for MongoDB by using mongoimport » Use NimoFullCheck to check data consistency after migration

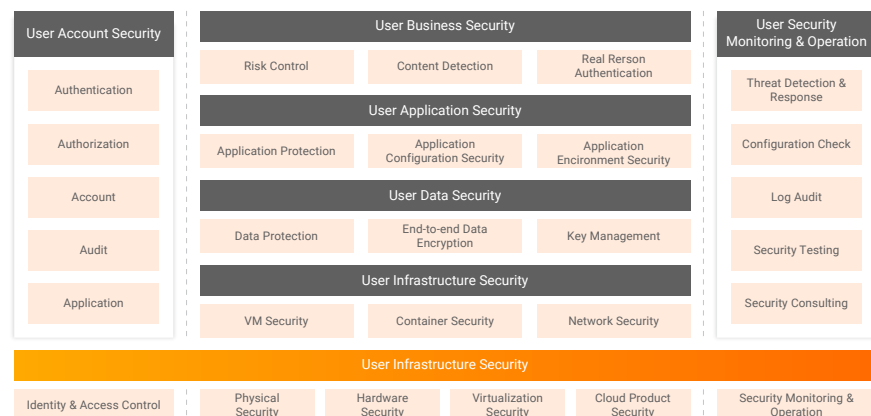
4.5.9 GREENPLUM

Source database on Azure	Target database on Alibaba Cloud	Migration guide
Self-built Greenplum on EC2	AnalyticDB PostgreSQL	» Migration guide



4.6 SECURITY

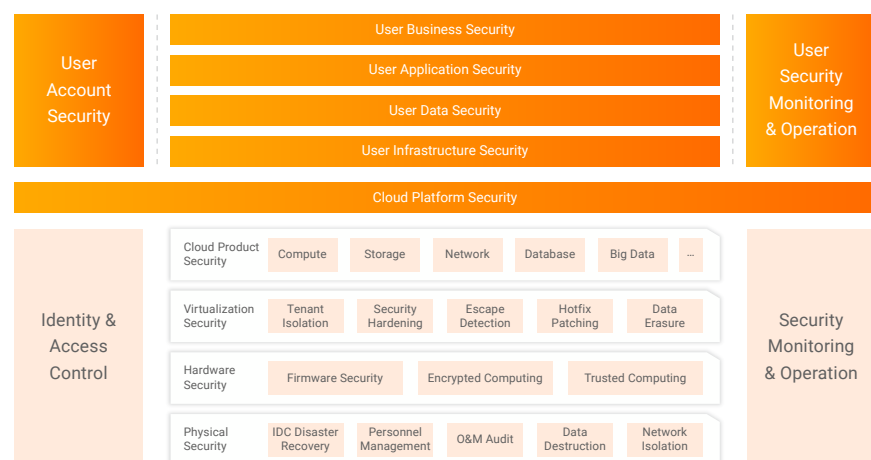
4.6.1 ALIBABA CLOUD SECURITY ARCHITECTURE



Alibaba Cloud provides a security architecture that offers protection in five horizontal dimensions and two vertical dimensions. The two vertical dimensions include account security (identity and access control) and security monitoring and operations. Note that these vertical dimensions include different implementations for both user security and cloud platform security. The five horizontal dimensions include cloud platform security at the bottom layer, as well as infrastructure security, data security, application security, and business security for cloud users.

This chapter introduces the overall security architecture and describes the key features of each layer that covers various Alibaba Cloud products and security services.

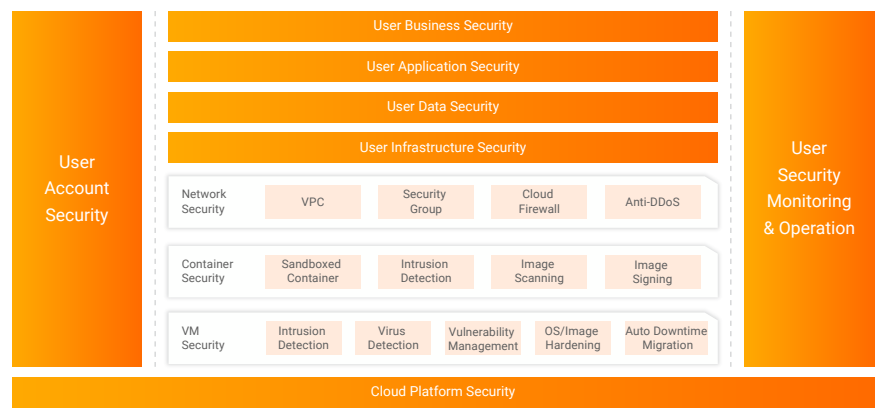
4.6.1.1 CLOUD PLATFORM SECURITY





The cloud platform security architecture includes the basic security capabilities provided by the Alibaba Cloud platform. In the two vertical dimensions, identity and access control and security monitoring and operations are invisible to customers. Similarly, customers can enjoy the high-level protection capabilities of Alibaba Cloud without performing configurations at the physical security, hardware security, and virtualization security layers. The cloud product security layer includes the security capabilities provided by the cloud platform for customers. Some capabilities such as tenant isolation are enabled by default. Some capabilities such as data encryption must be enabled and properly configured by customers to function properly.

4.6.1.2 USER INFRASTRUCTURE SECURITY

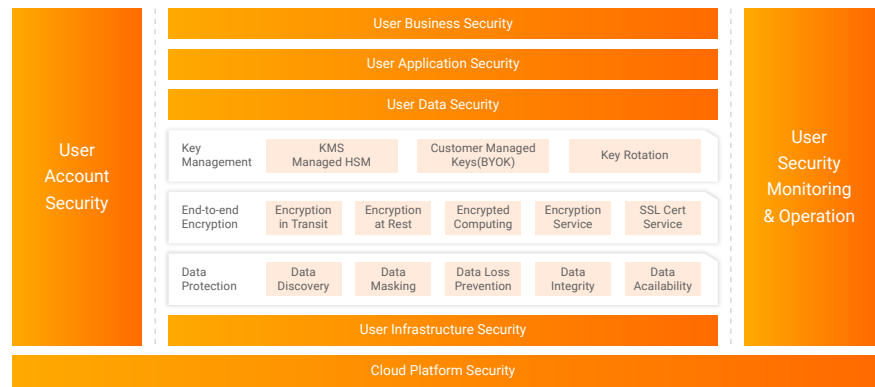


The user infrastructure security capabilities and requirements of cloud users focus on three aspects: host security, container security, and network security. These aspects involve the use of networks to isolate and protect the most important computing resources. Cloud applications and services of users are directly or indirectly built (through upper-layer cloud services) and run on the basic computing and network service modules.

In addition to computing and network resources, storage resources are also crucial to users, especially in the protection of user data. The user-side security architecture also includes two vertical dimensions: user account security and user security monitoring and operations. These vertical dimensions are involved in each horizontal dimension, such as configuration security checks for VM hosts during security monitoring and control over permissions of various cloud resources for account security.

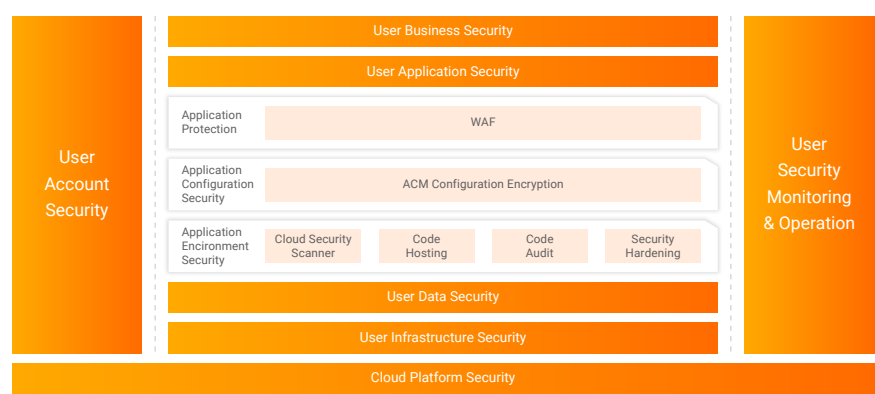


4.6.1.3 USER DATA SECURITY



User data security on the cloud is the lifeline of users and one of the most important representational attributes of the overall cloud security capability. In July 2015, Alibaba Cloud initiated the first Data Protection Proposal among cloud service providers in China. This public proposal appeals that the ownership of data of all developers, companies, governments, and social institutions on the cloud computing platforms belongs only to the users. Cloud computing platforms cannot use the data for other purposes. Cloud service providers have both the responsibility and obligation to help users ensure the confidentiality, integrity, and availability of their data.

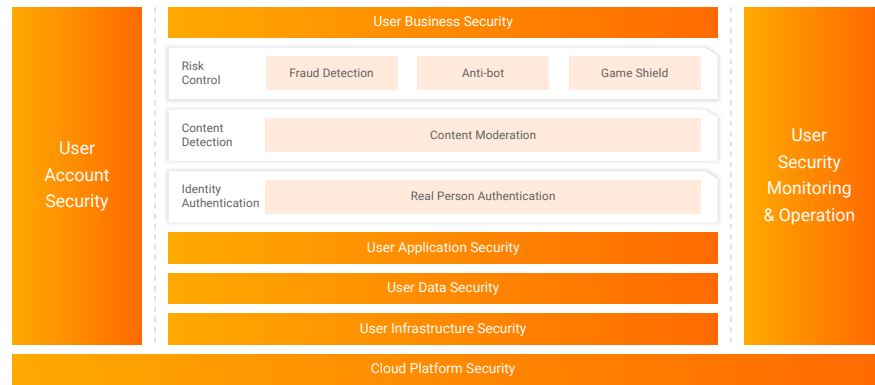
4.6.1.4 USER APPLICATION SECURITY



Applications built on Alibaba Cloud also need to be properly secured. At the application security level, Alibaba Cloud provides users with security capabilities in three aspects: application environment security, application configuration security, and application protection.

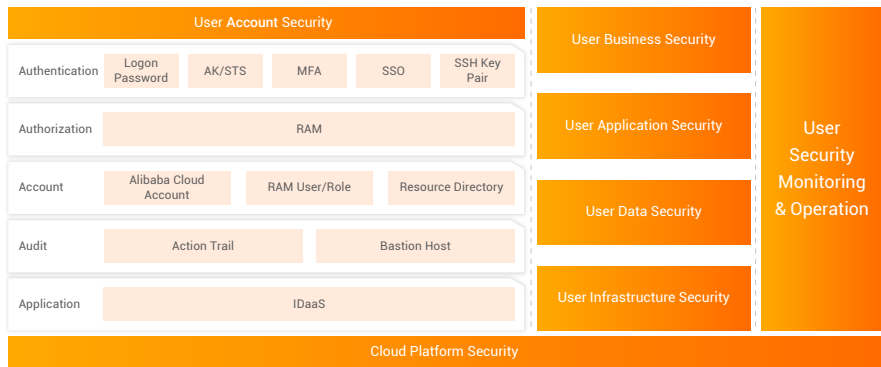


4.6.1.5 USER BUSINESS SECURITY



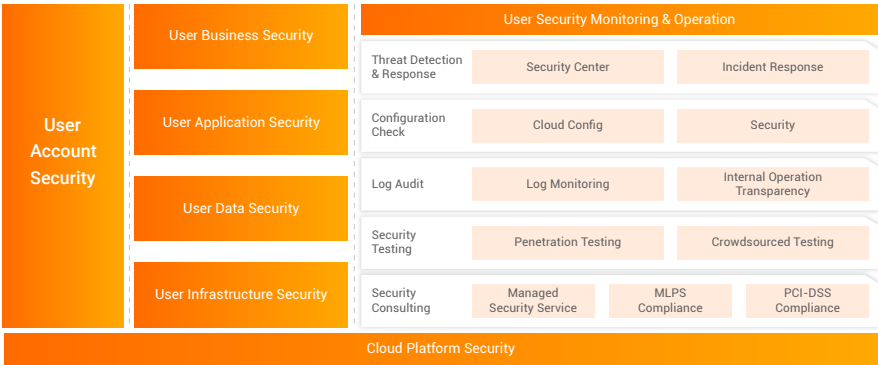
After a user's application has been built and runs on the Alibaba Cloud platform, Alibaba Cloud provides corresponding security capabilities for different business scenarios of the user. At the business security level, Alibaba Cloud provides security features from three aspects: identity authentication, content detection, and risk control.

4.6.1.6 USER ACCOUNT SECURITY



User account security is an important dimension in the overall cloud security architecture design. User account security on the cloud involves the “Five A”: Authentication, Authorization, Account Management, Audit, and Application Management. It is important to note that in cloud-based business, there is often a cloud product or service (such as RAM) that provides multiple dimensions of capabilities in the aforementioned aspects. This chapter describes Alibaba Cloud account security capabilities specific to each dimension.

4.6.1.7 USER SECURITY MONITORING AND OPERATIONS



After a user's resources, applications, business, and accounts are properly protected on the cloud, the cloud platform needs to provide security monitoring and operations capabilities to detect security threats, monitor the configurations of cloud resources and logs, and where appropriate, perform security testing on the user's overall business environment and provide a response and consulting services. In this way, the security monitoring and operations capabilities of Alibaba Cloud can effectively safeguard the user's business.

4.6.2 ANTI-DDOS

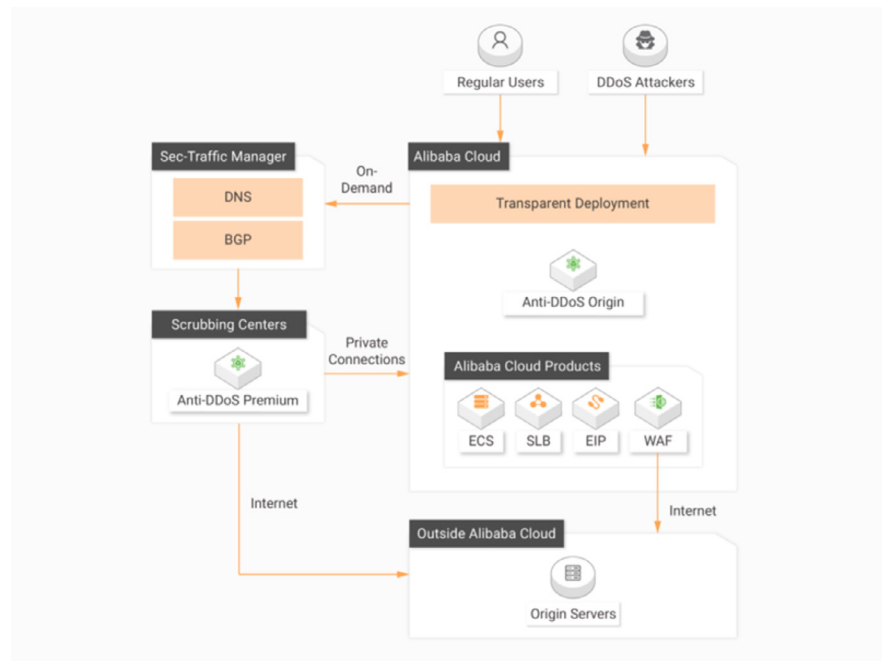
Anti-DDoS is provided for enterprises whose services on the internet become unavailable due to distributed denial of service (DDoS) attacks.

By default, Anti-DDoS provides an anti-DDoS capacity of up to 5 Gbit/s for free. In addition, Alibaba Cloud has launched a Security Credibility plan. After becoming a member of Security Credibility, you can enjoy the additional DDoS mitigation capacity on top of the default offering based on your security credibility score.



4.6.2.1 DDOS PROTECTION PACKAGE

The anti-DDoS protection package is designed for large-scale enterprises on Alibaba Cloud. It uses the native Alibaba Cloud network and transparent protection engine to mitigate DDoS attacks. The protection package can be used to protect various Alibaba Cloud products, such as Elastic Compute Service (ECS), Server Load Balancer (SLB), Web Application Firewall (WAF), and Elastic IP Address (EIP).



Procedure

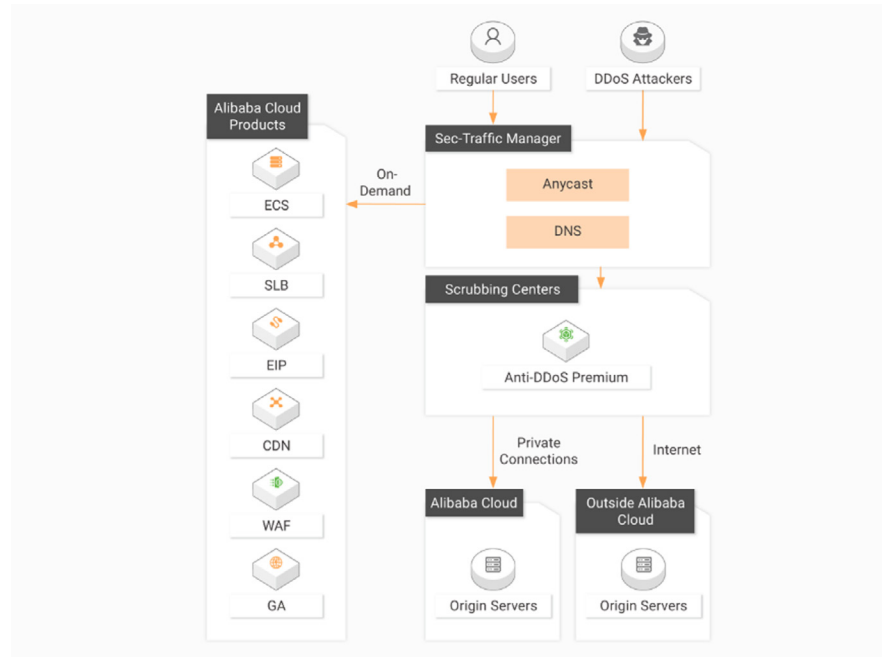
1. Purchase an Anti-DDoS Origin Enterprise instance

Note (Anti-DDoS Origin Enterprise instances are available by default only in mainland China. If you want to purchase an Anti-DDoS Origin Enterprise instance outside mainland China, you must submit a ticket or contact sales personnel. After the request is approved, you can purchase an Anti-DDoS Origin Enterprise instance).

2. Log on to the Anti-DDoS Basic console.
3. On the Anti-DDoS Origin page, locate the target instance and click 'Add a protection target' to add the addresses that you want to protect to the instance.

4.6.2.2 ANTI-DDOS PRO AND PREMIUM

Anti-DDoS Pro and Anti-DDoS Premium are designed for enterprises that deploy their resources on or off Alibaba Cloud. By using the massive traffic scrubbing center resources of Alibaba Cloud, both products work together with the artificial intelligence (AI) protection engine and adopt full-traffic proxy to protect against high-volume traffic attacks and refined web application level resource exhaustion attacks such as HTTP flood attacks.



4.6.2.3 SET UP ANTI-DDOS PRO USING DOMAINS

1. Add forwarding rules.

In the Anti-DDoS Pro or Anti-DDoS Premium console, add a website service that you want to protect by using a domain name, associate the service with an Anti-DDoS Pro or Anti-DDoS Premium instance, and configure the traffic forwarding rules.

2. Configure service traffic forwarding.

Modify the DNS records of your domain name to reroute the traffic directed to your website to an Anti-DDoS Pro or Anti-DDoS Premium instance. The instance scrubs the traffic and then forwards the traffic to the origin server, which protects your website service against DDoS attacks.

3. Configure protection policies.

After you set up an Anti-DDoS Pro or Anti-DDoS Premium instance to protect your website service, Intelligent Protection is enabled automatically. You can manually adjust anti-DDoS protection policies for your website service, which include



Intelligent Protection, Black Lists and White Lists (Domain Names), Blocked Regions (Domain Names), Accurate Access Control, and Frequency Control.

4. View the protection data of your website services

After you set up an Anti-DDoS Pro or Anti-DDoS Premium instance to protect your website service, you can use the Security Reports feature and log-related features to view the protection data in the Anti-DDoS Pro or Anti-DDoS Premium console.

4.6.2.4 SET UP ANTI-DDOS PRO USING IPS AND PORTS

1. Create a port forwarding rule.

Add a non-website service that you want to protect by using a port in the Anti-DDoS Pro or Anti-DDoS Premium console. Use the IP address of your Anti-DDoS Pro or Anti-DDoS Premium instance as your service IP address to reroute inbound traffic to your instance. After you change the IP address, the instance scrubs the inbound traffic and then forwards the traffic to the origin server.

2. Configure port forwarding and anti-DDoS protection policies.

Configure port forwarding policies as required, such as session persistence and health checks for multiple origin IP addresses. You can also configure anti-DDoS protection policies for non-website services, such as False Source, Speed Limit for Destination, Packet Length Limit, and Speed Limit for Source.

3. View the protection data of a port.

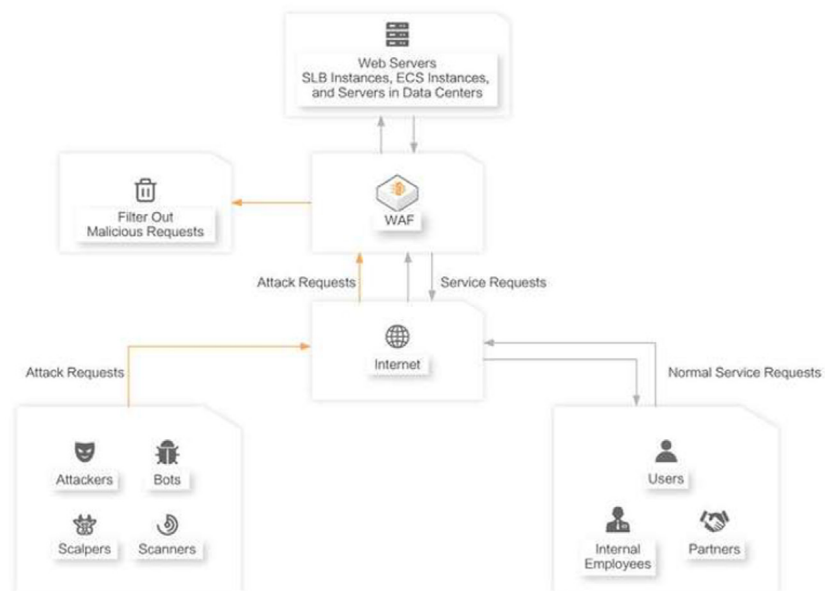
View the traffic that goes through a port on the Security Overview page of the Anti-DDoS Pro or Anti-DDoS Premium console after you set up an Anti-DDoS Pro or Anti-DDoS Premium instance to protect your non-website service.



4.6.3 WAF

Based on the big data and intelligent computing capabilities of Alibaba Cloud Security, Alibaba Cloud WAF defends against common security threats reported by OWASP. These threats include SQL injection, cross-site scripting (XSS), common vulnerabilities in web server plugins, web shell upload, and unauthorized access to cloud resources. WAF also filters out massive numbers of malicious access attempts, prevents leakage of web assets and data, and ensures the security and availability of web applications.

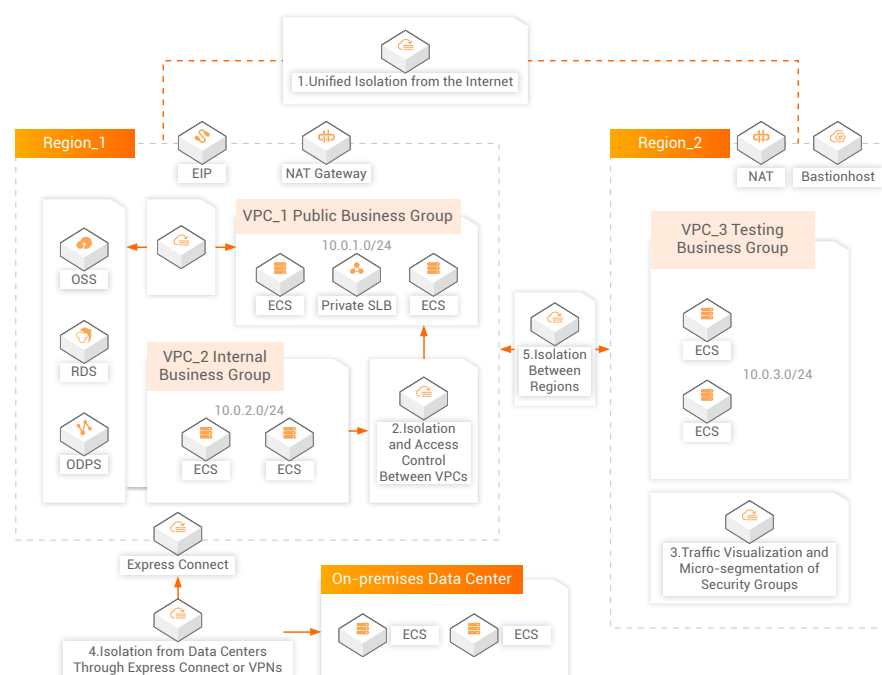
When you add your website to WAF in CNAME record mode, you must enter the website information and change the DNS record to resolve the domain name of your website to the CNAME assigned by WAF. This way, the requests destined for your website are redirected to WAF. This mode is supported regardless of whether your origin server is deployed on the cloud.



Alibaba Cloud offers Hybrid Cloud WAF to help protect and manage web applications in various hybrid cloud environments in a centralized manner. The web applications can be deployed across Alibaba Cloud Public Cloud and data centers and across Alibaba Cloud Public Cloud and third-party clouds. In this case, Hybrid Cloud WAF protects your service traffic that cannot be migrated to the cloud. To deliver an elastic and effective system, Hybrid Cloud WAF combines shared and exclusive resources in and outside the cloud. This way, you can protect the web applications in a centralized manner.

4.6.4 CLOUD FIREWALL

Cloud Firewall is the industry's first firewall as a service (FWaaS) solution for public cloud environments. This product manages the access control policy for north-south traffic from the Internet to ECS instances and the micro-isolation policy for east-west traffic between ECS instances in a centralized manner. Cloud Firewall comprehensively checks the exposure of your assets to the internet in the cloud environment and manages the access policies for public network IP addresses in a centralized manner. With one-click access, Cloud Firewall is the primary network security infrastructure for ensuring the network security of your businesses migrated to Alibaba Cloud. With the built-in intrusion prevention system (IPS), this product provides intelligent prevention, detects victim hosts, blocks external connections launched by hosts, and visualizes network-wide traffic and access relationships among ECS instances. Cloud Firewall meets the requirements of MLPS 2.0 for virtual boundaries, access control, IPS, and storage time of security logs, which must be retained for at least six months. This product is a necessary choice to be compliant with MLPS 2.0.

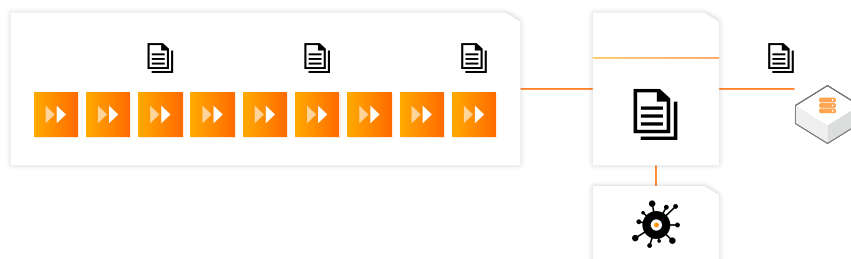


Procedure

1. On the Firewall Settings page, enable firewalls.
2. Choose Intrusion Prevention > Intrusion Prevention to configure intrusion prevention policies.
3. On the Access Control page, configure access control policies.
4. On the Traffic Analysis page, view traffic statistics. You can also view information about outbound connections.

4.6.5 SECURITY CENTER

Security Center is a unified security management system that identifies, analyzes, and alerts against security threats in real time. With security capabilities such as anti-ransomware, anti-virus, tamper-proofing, and compliance assessment, users can automate security operations, response, and threat tracing to safeguard cloud and local servers and meet regulatory compliance requirements.



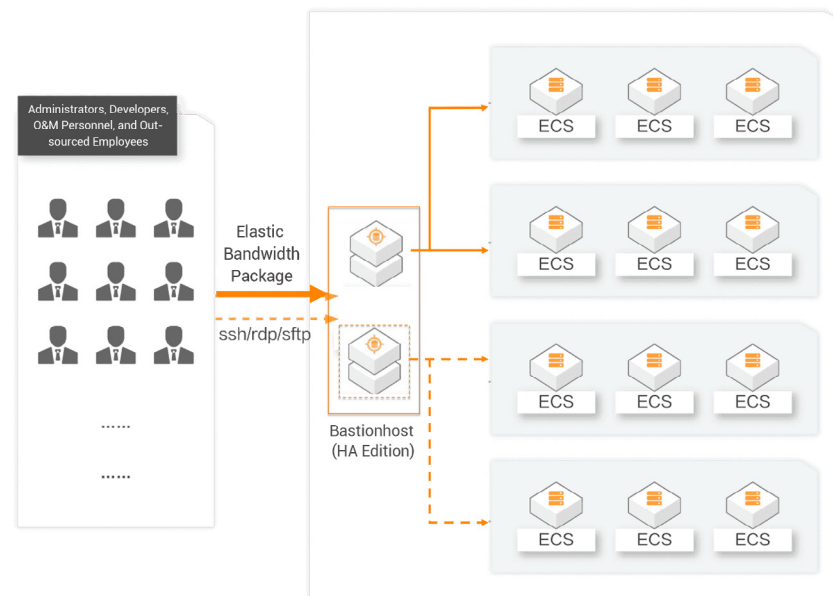
Procedure

1. Log on to the Security Center console.
2. In the Unprotected assets (ECS) section on the Overview tab, you can view the number of unprotected servers that do not have the Security Center agent installed.
3. Click Install now to go to the Settings page. Click the Agent tab and then the Client to install the tab. On the tab that appears, you can view the total number and list of servers that do not have the Security Center agent installed.
4. Click the Client Installation Guide tab to choose automatic installation, or download the installation package and run commands to install the Security Center agent.



4.6.6 BASTION HOST

Bastion Host provides multiple functions, such as O&M identity authentication, account management, and system operation audit. Bastion Host acts as a forward proxy and records the data flow of common O&M protocols, such as Secure Shell (SSH), Remote Desktop Protocol (RDP), and Secure File Transfer Protocol (SFTP). It can also regroup the data flows and reproduce them to implement an O&M audit.



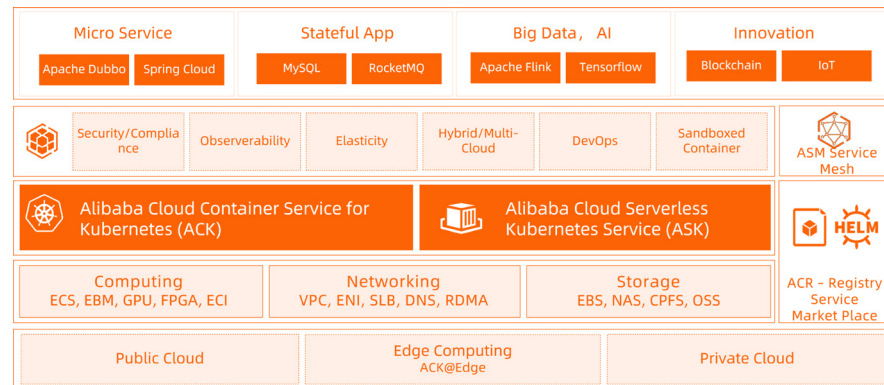
Procedure

1. Synchronize ECS assets.
2. Import Alibaba Cloud RAM users.
3. Create O&M rules.
4. Perform O&M operations on hosts.
5. Audit O&M sessions.

4.7 CONTAINER

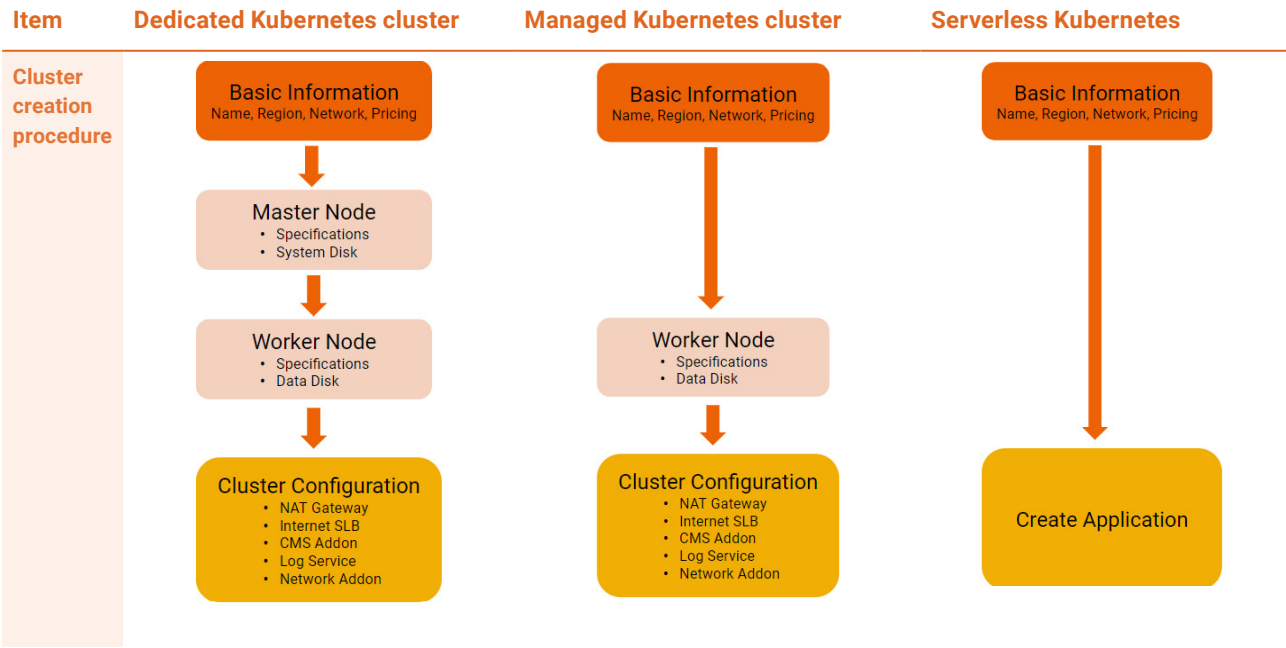
Container Service for Kubernetes (ACK) is one of the first services to pass the Certified Kubernetes Conformance Program. ACK provides high-performance management services for containerized applications. You can manage enterprise-level containerized applications throughout the application lifecycle.

The following figure shows the architecture of the Alibaba Cloud Container Service product portfolio.



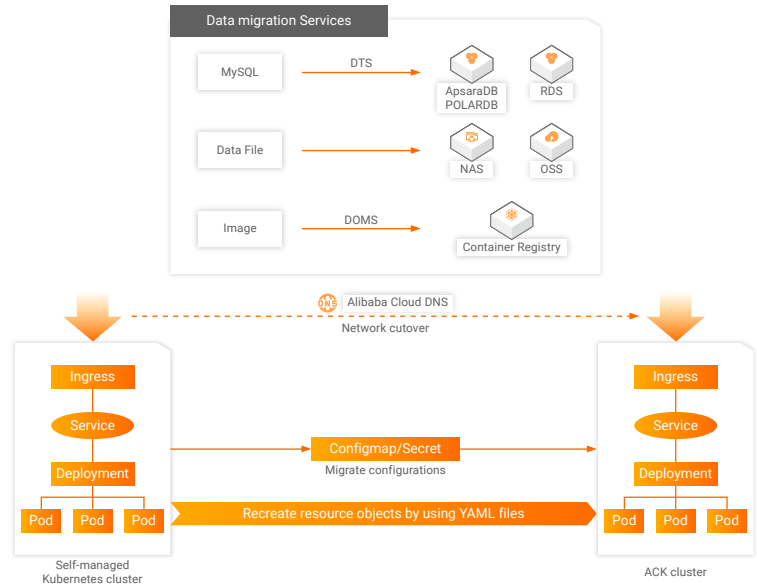
ACK provides the following three cluster types: dedicated Kubernetes clusters, managed Kubernetes clusters, and serverless Kubernetes clusters.

Item	Dedicated Kubernetes cluster	Managed Kubernetes cluster	Serverless Kubernetes
Feature	You must create master nodes and worker nodes.	You only need to create worker nodes. ACK creates and manages master nodes.	You do not need to create master or worker nodes.
	A dedicated Kubernetes cluster allows you to manage the cluster infrastructure in a more fine-grained manner. You must design, maintain, and upgrade the Kubernetes cluster on your own.	A managed Kubernetes cluster is easy-to-use, cost-efficient, and highly available. You do not need to manage master nodes.	A serverless Kubernetes cluster allows you to start applications directly. You do not need to manage nodes.
Billing method	You are not billed for cluster management features. However, you must pay for master nodes, worker nodes, and other infrastructure.	» Standard ACK clusters: You are not billed for cluster management features. However, you must pay for worker nodes and other infrastructure. » <u>Professional ACK clusters</u>: You are billed based on the number of clusters or the subscription method.	You are billed for the resources consumed by pods and the resource usage duration. The duration is measured in seconds.
Use scenario	Applies to all scenarios.	Applies to all scenarios.	Applies to batch tasks, urgent application scale-out, and continuous integration (CI) or continuous delivery (CD).
User profile			



How to migrate a managed Azure AKS Kubernetes cluster to Container Service for Kubernetes (ACK) without service disruption. As Azure AKS provides a standard Kubernetes cluster. We can treat it as a self-managed Kubernetes.

4.7.1 MIGRATION SCHEME



Procedure

1. Create and configure an ACK cluster.

O&M engineers create an ACK cluster and configure cluster resources. This step makes it easy for developers to migrate applications to Kubernetes. For more information, see [Create a dedicated Kubernetes cluster](#). The following parameters are required when you create the ACK cluster:

- » **Cluster type:** We recommend that you create a professional ACK managed cluster. The control plane of the professional ACK managed clusters is managed by ACK. You need to create and manage only worker nodes. You are charged only for the resources used by the worker nodes.
- » **Operating system:** Select an operating system based on your requirements:
 - By default, the CentOS 7.6 or Aliyun Linux 2.1903 operating system is selected. We recommend that you keep the default settings.
 - If the default system kernel cannot meet your requirements, you can use a custom image. For more information, see [Use a custom image to create an ACK cluster](#).
- » **VPC:** Select a virtual private cloud (VPC) and vSwitches for the ACK cluster.
- » **SNAT:** Configure Source Network Address Translation (SNAT) entries for the VPC.
- » **Public access:** Expose the API server of the ACK cluster with an Elastic IP address (EIP).
- » **Cloud Monitor agent:** Install the Cloud Monitor agent on the Elastic Compute Service (ECS) instances of the ACK cluster.
- » **Log Service:** Install and configure the Log Service agent in the ACK cluster.

2. Migrate data.

Migrate databases:

- a. Create an ApsaraDB Relational Data Service (RDS) instance.
- b. Configure an ApsaraDB RDS whitelist to allow only specified IP addresses to access the RDS instance.
- c. Configure a PrivateZone. Use PrivateZone to resolve the domain name of the database to the IP address of the RDS instance. This saves you the need to modify the database configuration on applications.
- d. Migrate the data in the MySQL database of the self-managed Kubernetes cluster. Use Data Transmission Service (DTS) to migrate the data from the MySQL database to the ApsaraDB RDS database in full, incremental, or two-way synchronization mode. For more information, see [Migrate data from a user-created MySQL database to an ApsaraDB RDS for MySQL instance](#).



Migrate data:

- a. Activate Object Storage Service (OSS).
- b. Create an OSS bucket.
- c. Migrate the data that is stored in the self-managed Kubernetes cluster. Use the `ossimport` tool to migrate the data in batches from an on-premises server or a third-party cloud storage service to OSS. For example, you can choose a third-party cloud storage service such as Amazon S3, Microsoft Azure, or Tencent Cloud Object Storage. For more information, see [ossimport – Overview](#).

Migrate images:

- a. Create a Container Registry repository.
- b. Set the credential that is used to access the created Container Registry repository.
- c. Migrate the images of the self-managed Kubernetes cluster. You can use the `image-syncer` tool to migrate the images of the self-managed Kubernetes cluster to the Container Registry repository. For more information, see [Use image-syncer to migrate container images](#).

3. Migrate application configurations.

O&M engineers or developers migrate the cluster or application configurations. For more information, see [Migrate applications from self-managed Kubernetes clusters to ACK clusters](#).

4. Perform regression tests.

Test engineers perform regression tests on the ACK cluster without interrupting the online business.

- » Configure the domain names for regression tests.
- » Test applications.
- » Check the logged application events.
- » Check the monitoring metrics of the applications.

5. Switch all production traffic to ACK.

O&M engineers modify the Domain Name System (DNS) configuration to switch traffic to the ACK cluster.

- » Use the DNS service: Modify the DNS configuration to switch the traffic.
- » Update the configurations or code of the client to switch the traffic.



6. Bring self-managed Kubernetes cluster offline.

O&M engineers check whether the ACK cluster can be accessed as expected. Then, O&M engineers bring the self-managed Kubernetes cluster offline.

- » Check whether the ACK cluster can receive and send traffic as expected.
- » Bring the self-managed Kubernetes cluster offline.
- » Clear the backup files stored in the OSS bucket that is created when you migrate applications.

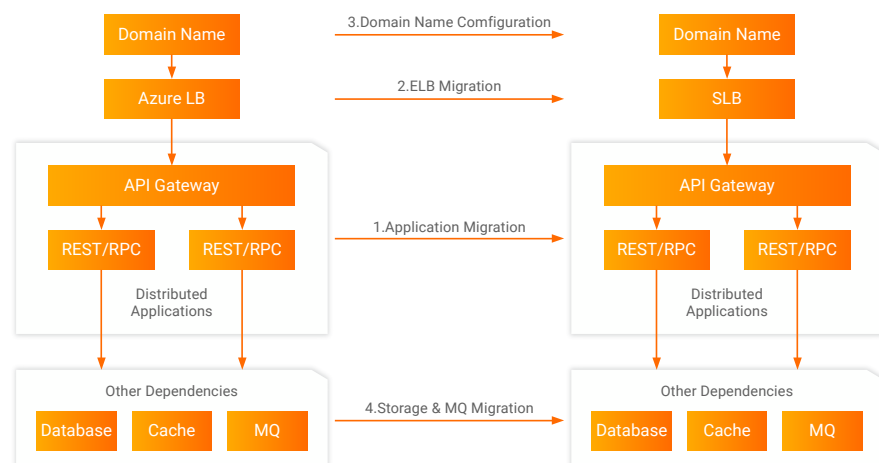
4.8 MICRO SERVICES

Enterprise Distributed Application Service (EDAS) is an Application Platform as a Service (aPaaS) offering that allows you to host applications and manage microservices. EDAS provides full-stack solutions to develop, deploy, monitor, and maintain applications and supports microservice frameworks such as Dubbo and Spring Cloud which is a good choice to migrate microservices-based applications from Azure to Alibaba Cloud.

Since your applications have been deployed to the production environment and are running in a normal state, it is the first priority to ensure the smooth operation of the business without interruption during the migration process. We have very good support for this scenario, which can ensure that the application platform runs uninterrupted during the migration process.

4.8.1 MIGRATION PROCEDURE

Specifically, we support two kinds of application migration of Spring Cloud and Dubbo microservice framework. And the migration process of these two types of applications is the same, mainly including the following four aspects:



1. Application Migration

Applications to be migrated are typically stateless and can be migrated first. For more detail, please refer to the following documents:

» [Spring Cloud](#)

2. LB Migration

- » If the Azure Load Balancer instance is used before migration, then after the application migration is complete, you need to migrate from LB to Server Load Balancer (SLB) instance on Alibaba Cloud.
- » If no Azure Load Balancer instance is used before migration, we recommend that you create and bind an SLB instance to the application after migration.

3. Domain Name Configuration

To create and bind a new SLB instance on Alibaba Cloud to migrated applications, you must add this SLB instance configuration to the domain name configuration. For more information, see [Change DNS servers for a domain name](#).

4. Storage and MQ Migration (Optional)

If your applications are deployed on Azure Spring App using other Azure services, such as Azure Database and Azure MQ, you still need to migrate storage and message queues along with applications. For this part, you can follow [Chapter 2.4](#) database migration.

4.8.2 MIGRATION SOLUTION

You can migrate applications by using two methods: registry switching, dual registration, and dual subscription. The two methods allow you to migrate applications during runtime without service discontinuity.

Microservice frameworks on Azure	Migration Solution	Description
Spring Cloud	Registry switching	Use Spring Cloud Alibaba to switch the original service registry to Nacos. Develop applications and deploy them to EDAS, and switch traffic by configuring the SLB and the domain name. Refer to this document for more details .
	Dual registration and dual subscription	Access the original service registry and the EDAS service registry when applications are migrated to enable mutual calling between migrated applications and non-migrated applications. Refer to this document for more details .



4.9 SERVERLESS

Alibaba Cloud Function Compute (FC) is a fully managed, event-driven compute service and is the best replacement for Azure Functions.

Function Compute allows customers to migrate codes without any change from Functions. It also integrates the container ecosystem. It provides image deployment and supports a seamless switch from the original container deployment method.

4.9.1 FUNCTION COMPUTE: CONTAINER IMAGE DEPLOYMENT

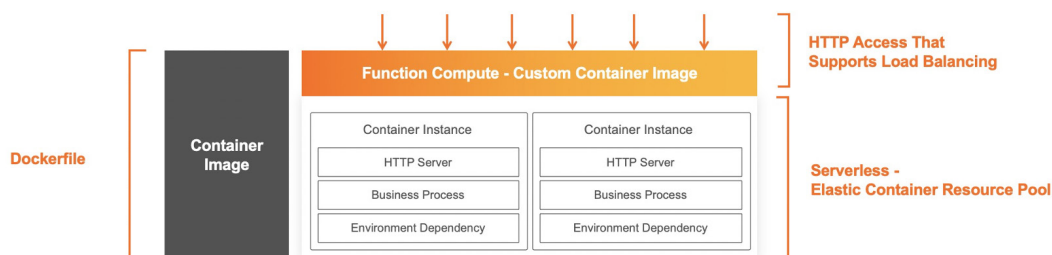
Function Compute integrates the container ecosystem. It provides image deployment and supports a seamless switch from the original container deployment method.

Disadvantages of conventional migration methods of standalone web applications (Spring Boot, WordPress, Flask, Express, and Rails):

- » O&M costs are increased to cater to modernized fine-grained responsibility division and service governance.
- » Historical assets such as on-cloud or off-cloud code, dependencies, and configurations make it difficult to go serverless.
- » You need to plan capacity and build O&M and monitoring systems.
- » Exclusive resources are allocated to low-traffic services, which causes low resource usage.

Advantages of application migration based on Function Compute and container images:

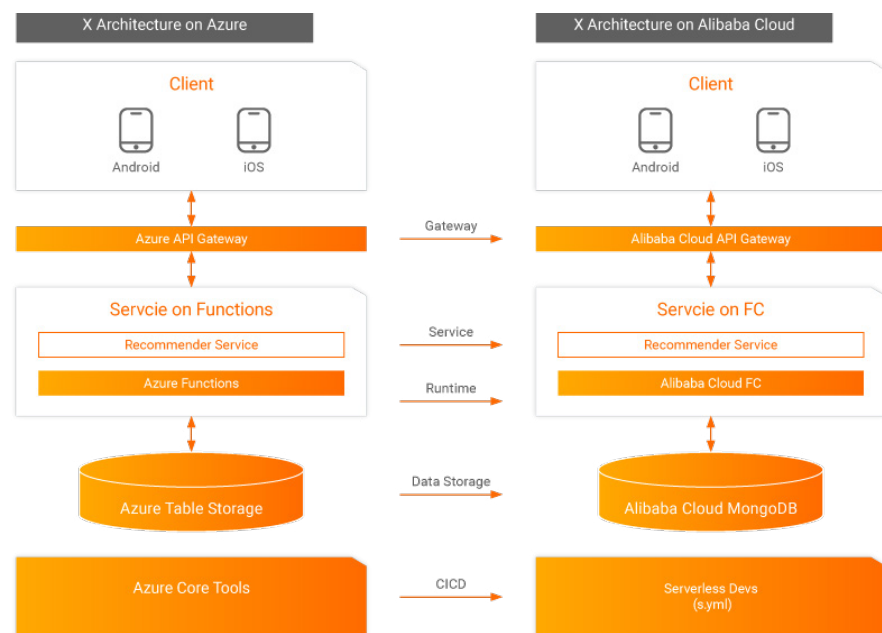
- » This method does not require O&M.
- » This method supports auto scaling. You do not need to arrange capacity.
- » This method ensures that all resources are in use, achieving
- » 100% resource usage and reducing IT costs.



Serverless integrates the capabilities of the cloud ecosystem based on SDKs and APIs. In particular, FC/Functions both integrate into the cloud ecosystem based on the unique event-driven callbacks.

As shown below, Alibaba has the most complete product matrix compared with Azure, so if the customer's services depend on other cloud products, the migration will be easy.

	Alibaba FC	Azure Functions	Tencent SCF
Storage	- OSS - NAS - SLS	- Azure Blob Storage - Azure Files	- COS - CLS
NoSQL DB	TableStore	Azure Table storage	- Not support
Messaging	- Kafka - MNS	Azure Queue Storage	Kafka
IoT	- IoT Platform - IoT Link Edge	Azure IoT Edge	- Not support
Event Hub	EventBridge	Azure Event Grid	- Not support
CDN Origin	Built-in Support	Built-in API Gateway	Via API Gateway
Monitoring & Operations	- Cloud Monitor - ARMS - Serverless Devs	- Azure Monitor - Azure Core Tools	- Not support
Orchestration	- Serverless Workflow - ROS	Durable Functions	Application Service Workflow (in Beta, no international version)
Outbound Integration	- EventBridge - FC - MNS Queue - MNS Topic	Event Grid	- Not support



4.10 MESSAGE QUEUES

4.10.1 MESSAGE SERVICE (MNS)

Alibaba Cloud Message Service is a distributed message queuing and notification service that supports concurrent operations to facilitate message transfer between applications and decoupled systems. Alibaba Cloud Message Service enables users to move data between distributed applications to achieve complex tasks, and build decoupled fault tolerant applications.



As a Pay-As-You-Go service, Message Service does not require any upfront investment and saves you the hassle of owning and scaling your own messaging platform. Best of all, the service is maintenance free.

Comparison between Message Queue for Message Service (MNS) and Azure Queue Storage

Category	Details	Message Service MNS	Azure Queue
Queue	Message Lifecycle	Supported Message state machine includes the following types: » Active » Inactive » DeletedExpired	Supported Message state machine includes the following types: » Active » Inactive » DeletedExpired
	Customizable message retention period	Supported	Supported
	Message delay (message scheduling time)	Supported	Supported
	Message visibility timeout	Supported	Supported
	Maximum message length configuration	Supported	Supported
	Dead-letter queue	Not supported	Supported
	FIFO queue (order/deduplication)	Not supported	Supported
Topic	Subscription filtering	Supported (Tag filtering)	Supported (JSON policy filtering)
	Queue subscription	Supported	Supported
	HTTP subscription	Supported	Supported
	SMS subscription	Supported	Supported
	Email subscription	Supported	Supported
	Mobile push	Supported	Supported
	Function Compute	Supported	Supported
	Dead-letter queue	Not supported	Not supported
	FIFO queue (order/deduplication)	Not supported	Not supported





Limit Comparison between Message Queue for Message service(MNS) and Azure Queue

Category	Details	Message Service MNS	Azure Queue
Queue Model	Queue Naming	Queue names are case-sensitive and can contain up to 120 characters.	Queue names are case-sensitive and can contain up to 80 characters.
	Message Retention Period	7 days	14 days
	Long Polling Interval	0s to 30s, default is 0s.	0s to 20s
	Maximum Payload Size	64 KB. To adjust the maximum payload size, a service request can be submitted.	256 KB
	Visibility Timeout	1s to 32,400s, default is 30s.	1s to 32,400s, with no default value.
	Queue Identification	Queue name: unique within the same region. MsgID: globally unique. ReceiptHandle: custom string.	Queue name: unique within the same region. MsgID: globally unique. ReceiptHandle: custom string.
	Message Delay	Set using the DelaySeconds parameter, with a range of 0s to 604,800s.	Set using the DelaySeconds parameter, with a range of 0s to 604,800s.
	Message Backlog Limit	Unlimited	Unlimited
	Single Queue QPS Limit	3,000. To adjust, a service request can be submitted.	Unlimited
	Message Traffic Limit	Maximum message size is 64 KB, with a traffic limit of 192 MB per second.	Unlimited
Topic Model	Topic Naming	120 characters, case-insensitive.	3 to 64 characters, case-sensitive.
	Maximum Message Length	64 KB	256 KB
	Retry Policy Limitations	Backoff mode: retry 3 times, with each retry interval randomly between 10s and 20s. Exponential decay mode: retry 176 times, with a total retry time of 1 day.	Backoff mode: retry 3 times, with each retry interval of 20s. Exponential decay mode: supports modifying the retry count.
	Single Topic Subscription Limit	100. To adjust, a service request can be submitted.	12,500,000
	Single Topic Access QPS Limit	500 QPS per topic	300 QPS per topic, or 10 MB per second, whichever is reached first.



Choose *Migrate Azure Queue to Message Queue for Message service (MNS)*

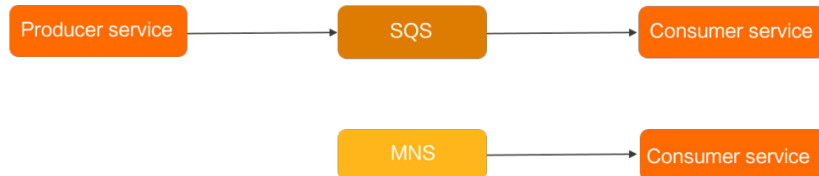
To ensure that your applications are smoothly migrated from Azure Queue to Alibaba Cloud MNS, we recommend that you synchronize queues, topics, and subscribed metadata between the services. Then, you must enable the dual-read and dual-write policies on the application side to migrate data. The following figure shows the process of how data is migrated from Azure Queue to Alibaba Cloud MNS.

Process:

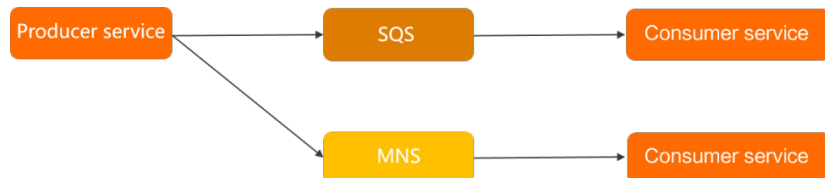
1. The existing workloads produce or consume messages based on Queue



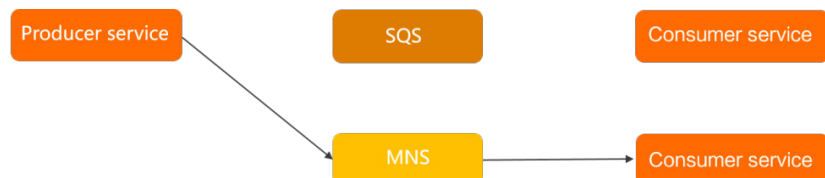
2. The consumer services simultaneously consume messages from Queue and Alibaba Cloud MNS.



3. The producer services are connected to Alibaba Cloud MNS.



4. The producer services are disconnected from Queue after the existing messages are consumed.



4.10.2 MESSAGE QUEUE FOR RABBITMQ

A distributed, fully managed, and professional messaging service that features high throughput, low latency, and high scalability. This service supports Advanced Message Queuing Protocol (AMQP) and is fully compatible with open-source RabbitMQ.

It also supports clients with different programming languages. The service is out-of-the-box and frees you from deployment and O&M. It allows you to easily migrate your business to the cloud while providing reliable and secure services.

Comparison between Message Queue for RabbitMQ and Azure MQ

Category	Item	Message Queue for RabbitMQ	Azure MQ
Performance	Peak transactions per second (TPS) of clusters	Not limited. Message Queue for RabbitMQ clusters adopts a distributed architecture that involves no primary nodes. This way, the number of nodes in use can be increased or decreased based on business needs, and thus the peak TPS of clusters is not limited.	Limited. To improve the performance of open-source RabbitMQ clusters, you must upgrade the specifications, which have upper limits.
	Peak TPS of a single queue	Not limited. Message Queue for RabbitMQ supports the increase of nodes in a single queue. No limit is imposed on the node concurrency, and a queue can have unlimited size.	Limited. The peak TPS of a single queue allowed is the peak TPS of a node allowed in the queue.
	Number of connections	Not limited. The more nodes a Message Queue for RabbitMQ instance has, the more connections the instance supports.	Limited. A single cluster supports only a limited number of connections, and the threshold cannot be adjusted to a higher value. This is due to the limit on cluster performance.
	Scheduled messages	Supports message scheduling that is accurate to seconds, high-performance, and out-of-the-box.	Supports message scheduling, which however is complex to use.
Stability	Message accumulation	Supports a large number of accumulated messages, and provides high performance even in scenarios of large message accumulation.	Cannot support a large number of accumulated messages. Breakdowns may occur due to memory overuse in scenarios of large message accumulation.
	Scalability	Adopts a distributed architecture that involves no primary nodes. The number of nodes in use can be increased or decreased with efficiency.	Requires you to modify the cluster specifications to adjust the cluster performance.
	Service availability	Adopts a distributed architecture that ensures the high availability of 99.95% of the clusters and provides high availability across zones.	Developed by using the Erlang programming language and requires advanced O&M experience. Availability cannot be ensured due to the open-source architecture.
	Data reliability	Uses the triplicate mechanism, which ensures high TPS performance.	Cannot ensure high TPS performance if you configure multiple replicas.
	Inspection system	Automatically detects and fixes issues such as deadlocks and breakdowns.	Not supported.

Category	Item	Message Queue for RabbitMQ	Azure MQ
General features	Support for client SDKs	Supports the open-source SDK for each language and each SDK version.	Supports open-source SDKs.
	Scheduled messages	Supports message scheduling that is accurate to seconds. You can use the x-delayed-message plug-in or specify the time-to-live (TTL) value to enable message scheduling.	Requires you to install a plug-in or transfer expired messages to support delayed messages.
	Transactional messages	Not supported.	Supported.
Exchange-related features	Exchange types	Supports direct, fanout, headers, topic, and x-delayed-message exchanges.	Supports direct, fanout, headers, topic, and x-delayed-message exchanges.
	Persistence	Supports both persistent and non-persistent storage of configurations.	Supports both persistent and non-persistent storage of configurations.
	Auto deletion	Supported.	Supported.
	Internal	Supported.	Supported.
	Alternate exchange	Not supported.	Supported.
Queue-related features	Queue type	Adopts a distributed architecture that requires no manual configuration of the queue type and provides high availability.	Requires manual configuration. Valid types: Classic: classic image queue. Quorum: quorum queue.
	Node	Requires no manual configuration and no O&M operations.	Requires manual configuration. You can select a node as needed.
	Persistence	Supports both persistent and non-persistent storage.	Supports both persistent and non-persistent storage.
	Maximum length	Requires no manual configuration and supports large message accumulation.	Requires manual configuration to prevent breakdowns caused by memory overuse in scenarios of large message accumulation.
	Maximum length (in bytes)		
	Maximum memory size		
	Maximum memory size (in bytes)		
	Retry limit	Requires no manual configuration. By default, a maximum of 16 retries is allowed. For more information about retries, see Message retry .	Requires manual configuration.
	Dead letter exchange	Supported.	Supported.
	Dead letter routing key	Supported.	Supported.
	Single active consumer	Not supported.	Supported.



Choose instance type for Message Queue for RabbitMQ

By opening the overview page in RabbitMQ Console, you can see how to choose an instance with the appropriate peak transactions per second (TPS).

Choose Migrate Azure? MQ to Message Queue for RabbitMQ

- » Migrate the metadata of the RabbitMQ cluster to the Message Queue for RabbitMQ instance, see [these detailed steps](#).
- » Migrate a RabbitMQ cluster to a Message Queue for RabbitMQ instance in dual-read mode:
 - » Enable a new consumer for the Message Queue for RabbitMQ instance to consume messages from the instance.
 - » Enable a new producer for the Message Queue for RabbitMQ instance, deprecate the original producer of the RabbitMQ cluster, and enable the original consumer to continue consuming messages from the cluster.
 - » After all the messages in the RabbitMQ cluster are consumed, deprecate the original consumer and the RabbitMQ cluster.

4.11 DEVOPS AND AUTOMATION (TERRAFORM)

Terraform is an infrastructure as code (IaC) tool that allows you to build, change, and version infrastructure safely and efficiently. This includes low-level components such as compute instances, storage, and networking, as well as high-level components such as DNS entries, SaaS features, etc. Terraform can manage both existing service providers and custom in-house solutions.

The official Alibaba Cloud team is responsible for the daily development and maintenance of Terraform Provider. If you have used AWS/Azure/GCP providers before, there will not be much change in the main framework, syntax, and usage habits. The main difference is the definition of products, resources, and business use. You can obtain detailed information about these contents through [Alibaba Cloud official documents](#).

The following table prepares the correspondence between commonly used products and resources for you so that you can quickly compare and find the corresponding document links.



Alibaba Cloud Product Name	TF Product/TF Resource Sample	Azure Product Name	TF Product/TF Resource Sample
Log Service	Log Service (SLS) / alicloud_log_project	» Azure Event Hubs » Azure Log Analytics » Azure Monitor	Messaging/ azurerm_eventhub LogAnalytics/ azurerm_log_analytics_cluster Monitor/ azurerm_monitor_action_group
Container Service for Kubernetes (ACK)	Container Service for Kubernetes (ACK) / alicloud_cs_autoscaling_config	Azure Kubernetes Service (AKS)	Container/ azurerm_kubernetes_cluster
Container Registry (ACR)	Container Registry (CR) /alicloud_cr_ee_instance	Azure Container Instances	Container/ azurerm_container_registry
Function Compute (FC)	Function Compute Service /alicloud_fc_alias	Azure Service Fabric Azure Functions	Service Fabric/ azurerm_service_fabric_cluster azurerm_function_app
AnalyticDB for MySQL	RDS /alicloud_db_instance	» Azure SQL Database » SQL Server Stretch Database	Database/ azurerm_mssql_database
ApsaraDB for Redis	RDS /alicloud_db_instance	» Azure Database for MySQL	Database/ azurerm_mssql_database
ApsaraDB RDS for MySQL	ECS /alicloud_instance	» Azure Database for PostgreSQL » Azure SQL Database Edge	Database/ azurerm_mssql_database
Elastic Compute Service (ECS)	ECS /alicloud_instance	Azure Bare Metal Servers (Large Instance Only for SAP Hana)	Compute/ azurerm_virtual_machine
Auto Scaling	Auto Scaling (ESS) /alicloud_ess_alarm	» Azure Autoscale » Virtual Machine Scale Sets	Automation/ azurerm_automation_account
Resource Orchestration Service (ROS)	ROS /alicloud_ros_template	Azure Resource Manager	Template/ azurerm_resource_deployment_script_azure_cli
Virtual Private Cloud (VPC)	VPC /alicloud_network_acl_attachment	Azure VNet	Network/ azurerm_network_interface
Server Load Balancer (SLB)	Classic Load Balancer (CLB) /alicloud_slb	Azure Load Balancer	Load Balancer/ azurerm_lb
VPN Gateway	VPN /alicloud_vpn_customer_gateway	Azure VPN Gateway	Network/ azurerm_vpn_gateway
Anti-DDoS	Anti-DDoS Pro /alicloud_ddosbgp_instance	Azure DDoS Protection	Network/ azurerm_network_ddos_protection_plan
Web Application Firewall (WAF)	Web Application Firewall (WAF) /alicloud_waf_instance	Azure WAF	Network/ azurerm_firewall



Alibaba Cloud Product Name	TF Product/TF Resource Sample	Azure Product Name	TF Product/TF Resource Sample
Resource Access Management (RAM)	RAM /alicloud_ram_access_key	Azure Active Directory	Active Directory Domain Services/ azurerms_active_directory_domain_service
Elastic Block Storage (EBS)	ECS /alicloud_disk	» Azure Page Blobs / Premium Storage » Managed Disks	Disks/ azurerms_disk_pool
Object Storage Service (OSS)	OSS /alicloud_oss_bucket	Azure Blob Storage	Storage/ azurerms_storage_blob
File Storage NAS	Network Attached Storage (NAS) / alicloud_nas_access_group	Azure Files	Storage/ azurerms_storage_share_file

For more information about Terraform hands-on, please refer to:
<https://github.com/alibabacloud-howto/terraform-templates>



ABOUT

Established in 2009, Alibaba Cloud is a global leader in cloud computing and artificial intelligence (AI). It is the technological backbone of the Alibaba Group, which aims to make it easy to do business anywhere. As one of the significant business units of the group, Alibaba Cloud provides the best-in-class cloud computing infrastructure and cloud services for global enterprises across industries to operate more efficiently, effectively, and securely. Alibaba Cloud also supports Alibaba Group's groundbreaking Double 11 Global Shopping Festival, which generated USD \$74.1 billion (RMB ¥498.2 billion) in gross merchandise value (GMV) in 2020 with zero downtime.

Headquartered in Hangzhou, China, the Alibaba Group provides vital technology infrastructure and marketing capabilities to help businesses grow their products and services online. The Alibaba Group encompasses commerce, cloud computing, digital media, and innovation. In addition to Alibaba Cloud – the cloud computing division of the Group – other business units include AliExpress, the global consumer marketplace, and Alipay, the mobile and online payment platform.

In 2017, Alibaba established the DAMO Academy (Academy for Discovery, Adventure, Momentum and Outlook), dedicated to scientific and technological research and innovation. Alibaba has now invested billions of dollars in research and development, and has built research labs in seven cities worldwide, looking into topics including data intelligence, FinTech, quantum computing, IoT, and human-machine interaction.

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